STAT 145 Course Pack

UWL Department of Mathematics & Statistics

1.1 Introduction to Data

Scientists seek to answer questions using rigorous methods and careful observations. These observations — collected from the likes of field notes, surveys, and experiments — form the backbone of a statistical investigation and are called ***data****. Statistics is the study of how best to collect, analyze, and draw conclusions from data. In this first chapter, we focus on both the properties of data and on the collection of data. We will introduce the building blocks of statistical thinking: observations, variables, data frames, and the types of questions that statistics can help us answer.*

Key Concepts

- Recognize that a data set consists of cases and variables
- Identify variables as either categorical or quantitative
- Determine explanatory and response variables where appropriate
- Determine whether a study is observational or experimental

Why we collect data

All statistical analyses start with a question or an idea. This question or idea is posed about a *population*, in particular a population *parameter*. The main purpose of collecting data is to learn about this unknown population parameter. We often cannot measure data for every individual in a population because it would cost too much money or take too long, so a *sample*, or subset of the population, is collected instead. Using the information in the sample, a *statistic* is calculated and used to estimate the corresponding population parameter. This process of using a sample statistic to learn about a population parameter is called *statistical inference*. One of the main goals of this class is to introduce you to the procedures for making statistical inferences in a variety of settings.

A *population* is a well-defined collection of *all* individuals or objects of interest.

A *parameter* is a number that describes a population and is almost always unknown.

A *sample* is a subset of the population on which we measure data.

A *statistic* is a number that describes a sample and is almost always known.

Inference is the process of drawing conclusions about a *population parameter* from a *sample statistic*.

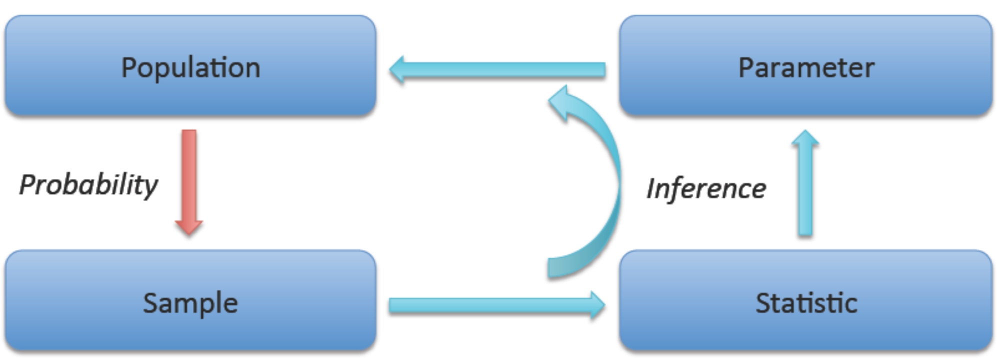

Figure 1: The cycle of statistical inference.

Data Basics

After data is gathered, it needs a place to be stored. In statistics, this is called a *data set*, or *data frame*. Data sets can be as simple as an Excel spreadsheet or as complicated as a major database. It all depends on the complexity of the question.

A *data set*, or *data frame* is used to store the set of measurements taken on individual units.

Structure: Cases and variables

In general, when data is stored in a table, each row of the table represents an *observation*, or a *case*, in the dataset. Each column of the table represents a *variable* in the dataset.

Note: Many datasets use the first column as an identifier. This would not be considered a variable.

Observations, or *Cases*, are the subjects/objects that we obtain information about. These are most generally known as *experimental units*.

A *Variable* is a characteristic that is measured and recorded for each case.

Class Example 1.1.1 *Distinguishing between cases and variables*

Let's get to know your classmastes. Find two people near you and ask them the following questions:

a) What is your first name?
b) What is your major?
c) How tall are you?
d) How many credit hours are you taking this semester?
e) How often do you drink caffeine (never, once a week, few times a week, everyday)?

Name	**Major**	**Height**	**Credit Hours**	**Caffeine**

How many cases are there?

How many variables are there?

Types of Variables

Before we can know how to answer a question, we need to think about the kinds of variables we have recorded. Variables that split cases into groups, called *levels*, are called *categorical*, or *qualitative*, variables. Variables that are numerical quantities are called *quantitative* variables.

Categorical variables can further be subdivided:

- *Nominal variables* have levels with no natural ordering. The variable *major* from Example 1.1 is nominal, there is no reason to rank one major over another.
- *Ordinal variables* have levels with a meaningful order. The variable *caffeine* has a natural ordering.

Categorical, or *Qualitative* variables divide the cases into groups, placing each case into exactly one of two or more non-numerical, non-overlapping categories.

Quantitative variables take on numerical values for each case, often by measuring or counting. Numerical operations like adding and averaging make sense for quantitative variables.

Quantitative variables can further be subdivided:

- *Continuous variables* can take any value within a range, including decimals. The variable *height* is continous.
- *Discrete variables* can only take counting values (typically whole numbers: 0, 1, 2, ...). The variable *credit hours* is discrete.

Note: In most of this course, we will treat categorical variables as nominal (unordered) and quantitative variables as continuous. The distinction between nominal and ordinal, and between discrete and continuous becomes more important in advanced courses when special methods are used to take advantage of the ordering.

The ability to identify whether a variable is categorical or quantitative is important because different types of variables are summarized in different ways.

Class Example 1.1.2 *Computer ownership at UWL*

UWL would like to estimate the proportion of students who own a computer. The university asks 500 students.

a) Identify the cases and variable(s) recorded.

b) What is the sample?

c) What is the population?

d) Is the variable categorical or quantitative?

e) Identify an unmeasured quantitative variable of interest.

f) Is the result of this study a statistic or parameter?

Explanatory and Response Variables

In this class, we will discuss two ways in which data are used to answer questions. The first is looking only at a single variable, and the second is to determine if there is a relationship between two variables. When two variables show some connection or pattern with one another, they are called *associated* variables (or related variables). If two variables are not associated, there is no evident relationship between them. The two variables are said to be *independent*.

In studies that look for an association, many times we want to know if one variable explains the behavior of another variable. In this kind of study, the *explanatory* variable would explain the behavior in the *response* variable.

Associated variables show some connection or pattern with one another.

Independent variables have no evident relationship between them.

An *explanatory* variable is one that helps us understand or predict the value of another variable.

The *response* variable is the variable that we are trying to understand.

Class Example 1.1.3 *Explanatory and response variables*

For the following research questions, identify the explanatory variable and the response variable.

a) Does eating yogurt cause people to lose weight?

b) Does meditation help reduce stress?

c) Is hyperactivity in children affected by sugar consumption?

Class Example 1.1.4 *2023 Hollywood Movies*

Here is a small part of a dataset that includes information on all 136 movies to come out of Hollywood in 2023. "Rating" is audience rating on a 100-point scale, "Budget" is budget in millions of dollars, and "Opening" is opening weekend gross, in millions of dollars.

Title	**Lead Studio**	**Rating**	**Genre**	**Budget**	**Opening**
Oppenheimer	Atlas	93	Biography	$100	$82
Barbie	Warner Bros	88	Comedy	$145	$162
Spider-Man: Across the Spider-Verse	Sony	95	Action	$100	$120.5
Guardians of the Galaxy Vol. 3	Disney	82	Sci-fi	$250	$118
The Holdovers	Focus	97	Drama	$13	$2.7
Killers of the Flower Moon	Paramount	93	Drama	$200	$23

a) What are the cases in this dataset?

b) What are the variables in this dataset? For each variable, identify whether it is categorical or quantitative.

c) Name a question we might ask about this dataset that is about

- a single variable

- a relationship between two of the variables

Observational Studies and Experiments

There are two primary types of data collection: experiments and observational studies. Understanding the difference is critical, because the type of study determines what conclusions we can draw.

If the researcher is only observing what happened, the study is an *observational study*. In statistics, the word observational does not mean that the researcher is watching the participant do every part of the study. Instead, observational refers to the fact that the researcher takes a dataset and makes observations about the data in the set. If the researcher is actively involved with assigning different values of the explanatory variable to the participants, then the study is an *experiment*. In general, observational studies can provide evidence of a naturally occurring *association* between variables, but they cannot by themselves show a *causal* connection. This is because in observational studies, researchers cannot control for all the other variables that might be influencing the outcome.

In an *experiment* the researcher actively controls one or more of the explanatory variables.

In an *observational study* the researcher simply observes the values as they naturally exist.

Two variables are *associated* if values of one variable tend to be related to the values of the other variable.

Two variables are *causally associated* if changing the value of one variable influences the value of the other variable.

Warning: Association does not imply causation.

Class Example 1.1.5 *Experimental or Observational Studies*

Determine whether each of the following is an experiment or an observational study:

a) A pharmaceutical company randomly assigns 200 patients to receive either a new drug or a placebo and measures their cholesterol levels after 6 months.

b) A university surveys its graduates to determine whether those who participated in study-abroad programs earn higher salaries five years after graduation.

c) A teacher assigns students in her morning class to use a new learning app and students in her afternoon class to study using traditional methods, then compares exam scores.

Summary

This chapter introduced you to the world of data. Here are the key ideas:

- *Data* are observations collected about the world around us. *Statistics* is the science of collecting, analyzing, and drawing conclusions from data.
- A *data set*, or *data frame*, organizes data so that each row represents an *observation* and each column represents a *variable*.
- Variables come in two fundamental types: *quantitative* (continuous or discrete) and *categorical* (nominal or ordinal).
- The *explanatory* variable is the variable we think might influence the *response* variable.
- *Experiments* randomly assign subjects to treatments and can establish causation. *Observational* studies merely observe what happens and can identify associations, but generally cannot establish causation.
- Association does not imply causation — this is one of the most important principles in statistics.

Many of the ideas from this chapter will be revisited as we move on to doing end-to-end data analyses. In the next chapter, you will learn about how we can design studies to collect the data we need to make conclusions with the desired scope of inference.

1.2 *Data Collection and Study Design*

Before digging into the details of working with data, we pause to think about how data come to be. If data are to be used to draw broad conclusions, then it is important to understand who or what the data represent. One important aspect is ***sampling*** *— knowing how the observational units were selected from a larger group allows us to generalize back to the population from which the data were drawn. Additionally, by understanding the structure of the study, we can separate causal relationships from mere associations. A good question to ask before analyzing any dataset is:* ***How were these observations collected?*** *You will learn a lot about the data by understanding its source.*

Key Concepts

- Distinguish between a sample and a population
- Recognize when it is appropriate to use sample data to make inferences about the population
- Critically examine the way a sample is selected, identifying possible sources of bias
- Recognize that random sampling is a powerful way to avoid sampling bias
- Identify other potential sources of bias that may arise in studies on humans
- Distinguish between an observational study and a randomized experiment
- Recognize that only randomized experiments can lead to claims of causation
- Distinguish between a randomized comparative experiment and a matched pairs experiment
- Identify potential confounding variables in a study

Motivation

A state in the upper Midwest is currently considering a bill that would provide amnesty to underage persons who call for help from the authorities with an overly intoxicated person. One of the state senators who represents a district that contains a university is hesitant to initially support the bill, so she has commissioned a study to learn about how the residents of her district feel about the bill. The senator will select from one of the three study options.

Study 1: "Would you support a bill that prevents authorities from giving a ticket to an underage person who has asked for help with an overly intoxicated person?" Participants would be recruited by purchasing ads on the local radio stations.

Study 2: "Would you support a bill that has the potential to allow underage people to drink with no fear of retribution?" Participants would be a random sample of all taxpayers in the district gathered by door-to-door interviews.

Study 3: "Would you support a bill that helps reduce the chances of undergraduate death due to alcohol poisoning by guaranteeing that underage people who request emergency medical assistance for someone who is very intoxicated will not receive disciplinary sanctions from the university?" Participants would be a random sample of all property owners in the district contacted to fill out a survey via email.

a) Rank the three studies based on the question they ask. Briefly justify your ranking.

b) Rank the three studies based on the sampling method. Briefly justify your ranking.

c) If you could mix the questions and sampling methods, which combination do you think would be the best?

Populations and Samples

The first step in conducting research is to identify the question to be investigated. A clearly stated research question helps identify what subjects or cases should be studied and what variables are important, or the *population*. It is also important to consider how data are collected so that they are reliable and help achieve the research goals.

A *population* includes **all** individuals or objects of interest.

A *census* occurs when every case in the population is measured.

The *sampling frame* is a list of the names of all subjects or objects in the population.

Data are collected from a *sample*, which is a subset of the population.

Statistical inference is the process of using data from a sample to gain information about the population.

One way we could find an answer is to look at every case by taking a *census*. However, most of the time this is not feasible. Taking a census could take months and is often very costly. Sometimes items are destroyed in the sampling process, such as measuring the tensile strength of a wire or the length of life of a light bulb or battery. Collecting a census in these cases would be foolish. So, most of the time, researchers take a *sample* from the population and use what they find from the sample to describe the population

In order to make sure that all units in the population are included, a *sampling frame* is compiled and used to select the subjects for the sample. This process of using a sample to describe the population is called *statistical inference*, and the accuracy of the inference can be greatly impacted by the quality of the sample and how well the sample reflects the characteristics of the population.

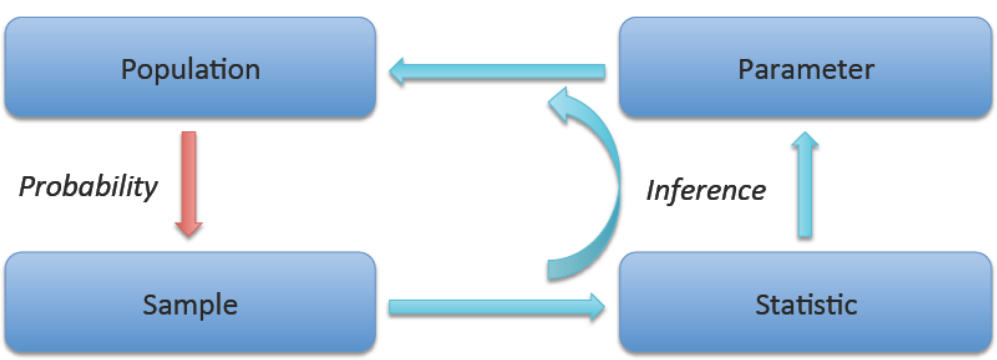

Figure 2: The cycle of statistical inference.

Class Example 1.2.1 *Population Identification*

For the following research questions, identify the target population and what represents an individual case

a) What is the average mercury content in swordfish in the Atlantic Ocean?

b) Over the last five years, what is the average time to complete a degree for UW–La Crosse undergrads?

c) Does a new drug reduce the number of deaths in patients with severe heart disease?

Parameters and Statistics

In most statistical analyses, the research question boils down to understanding a numerical summary — perhaps a quantity you already know (like the average) or one you will learn about in this course.

A numerical summary can be calculated on either the sample, called a *sample statistic*, or the entire population, called a *population parameter*. However, measuring every unit in the population is usually impossible. So we calculate the summary statistic from a sample and use it to estimate the corresponding population value.

A *parameter* is a number that describes a population and is almost always unknown.

A *statistic* is a number that describes a sample and is almost always known.

Class Example 1.2.2 *Parameter and Statistic Idenification*

For the following research questions, identify the population parameter and the sample statistic.

a) A sample of 60 swordfish have an average of 0.97 ppm

b) An average of 4.37 years is reported from 25 UWL students as the time to complete their degree.

c) After 1000 patients started taking a new drug for severe heart disease, 28% were reported to have died within 5 years.

Sampling from a Population

We might try to estimate the time to graduation for UW–La Crosse undergraduates by collecting a sample of graduates. All graduates in the last five years represent the *population*, and graduates who are selected for review are collectively called the *sample*. In general, we always seek to *randomly* select a sample from a population. The most basic type of random selection is equivalent to how raffles are conducted. For example, we could write each graduate's name on a raffle ticket and draw 10 tickets. The selected names would represent a random sample of 10 graduates.

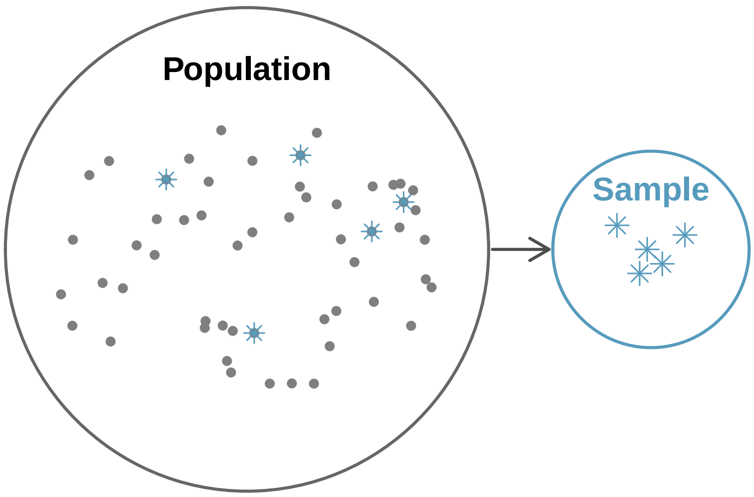

Figure 3: Sampling from a population. Individuals are randomly selected from the population (large circle) to be included in the sample (small circle).

If we ask a student who happens to be majoring in nutrition to select several graduates for the study, they might pick a disproportionate number of graduates from health-related fields. When selecting samples by hand, we run the risk of picking a *biased* sample, even if our bias is unintentional.

Bias occurs when the results from the sample differ from the population.

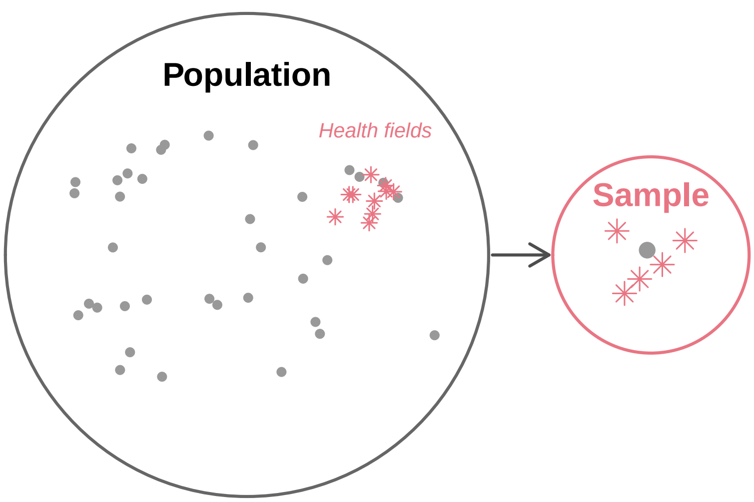

Figure 4: A biased sample. A nutrition major asked to select graduates might inadvertently pick a disproportionate number from health-related fields.

If someone is permitted to pick and choose exactly which individuals are included in the sample, it is entirely possible that the sample will overrepresent that person's interests — even if the bias is unintentional. Sampling randomly helps address this problem.

The act of taking a simple random sample helps minimize bias. However, bias can crop up in other ways. For example, asking sensitive questions may result in *response bias* because participants do not feel comfortable answering the question truthfully. If a lot of people do not respond to a survey, it will suffer from *non-response bias*. Finally, if the population from which the sample is drawn excludes groups of people, then the sample will suffer from *undercoverage*.

Response bias is the tendency for people to not answer questions honestly.

Non-response bias occurs when the response rate is very low.

Undercoverage occurs when part of the population is not considered for inclusion in the sample.

Class Example 1.2.3 *Wording bias*

A survey is to be conducted using a random sample of citizens in a town, asking them if they support raising taxes to increase funding for the public school.

a) Write the question in a way that is likely to bias the results toward more *yes* answers.

b) Write the question in a way that is likely to bias the results toward more *no* answers.

Sampling Methods

Good Ways to Obtain a Sample

There are a number of appropriate ways to randomly select your sample. The most basic type of sampling is the *simple random sample* (*SRS*). Taking a SRS avoids sampling bias and tends to produce a good representaitive sample of the population. However, there are many times when we can't conduct a simple random sample. *Stratified random sampling* occurs when the population is divided into homogeneous, non-overlapping groups (called *strata*) and a simple random sample is selected from each group. *Cluster random sampling* occurs when the population is divided into many non-overlapping groups (called *clusters*), a simple random sample of clusters is obtained, and data is collected from every case in the cluster.

A *simple random sample* is obtained when every sample of a given size is equally likely to be the one selected.

Stratified random sampling occurs when the population is divided into homogeneous, non-overlapping groups (called *strata*) and a simple random sample is selected from each group.

Cluster random sampling divides the population into many non-overlapping groups (called *clusters*), randomly selects a subset of clusters, and collects data from every case in the cluster.

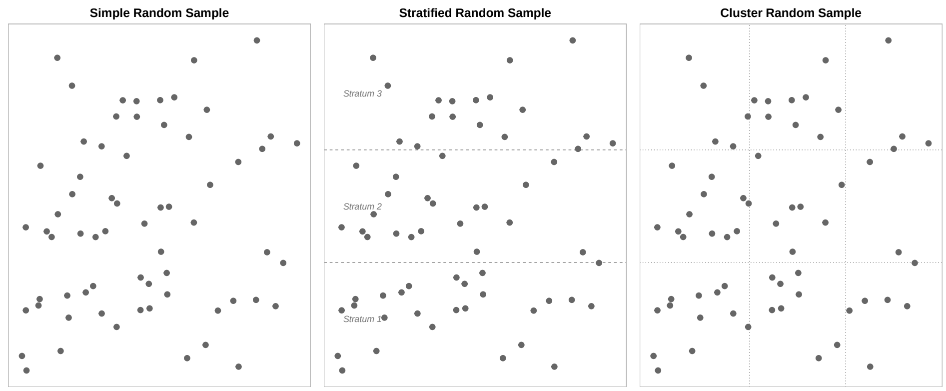

Figure 5: Three different ways we can collect a random sample. Population shown as dots — circle sampled individuals during lecture.

Class Example 1.2.4 *Picking a good sampling method*

Which type of sampling method is most appropriate in the following scenarios?

a) Researchers want to test a new math curriculum for third graders. They only have the budget to try the curriculum in 10 elementary schools.

b) Processor chips are continuously manufactured at a production plant. To ensure that they meet size specifications so that they will fit inside an iPhone, manufacturers want to test 100 chips throughout the day.

c) A player's agent wants to know the average salary for each position of major league baseball.

Table 1: Sampling method summary

Method	How it works	Best used when...
SRS	Every case has an equal chance of selection	Population is relatively homogeneous and accessible
Stratified	Divide into similar groups (strata), then SRS within each	Known subgroups differ on the outcome of interest
Cluster	Randomly select entire groups (clusters)	Clusters are internally diverse but similar to each other; travel/cost is a concern
Multistage	Randomly select clusters, then SRS within each	Same as cluster, but clusters are large

Random sampling caution: Random is not the same as haphazard. We cannot obtain a random sample by haphazardly picking a sample on our own. We must use a formal random sampling method such as technology or drawing names out of a hat.

Class Activity: Sampling from Gettysburg Address Worksheet

Bad Ways to Take a Sample

Without a random sample, we may encounter *sampling bias* which occurs when the sample does not accurately represent the population of interest. One common sampling technique that will nearly always be biased is *voluntary response sampling*. With this kind of sampling, a researcher asks people to volunteer to be a part of a study. Another common type of sampling that produces biased results is the *convenience sample*. In a convenience sample, a researcher simply uses the most convenient group available as the sample.

Sampling bias occurs when the sample does not accurately represent the population of interest.

Voluntary response sampling occurs when a researcher asks people to volunteer to be participants instead of randomly selecting possible participants.

Convenience sampling happens when a researcher just uses the most convenient group as a sample.

Class Example 1.2.5 *Driving with a pet on your lap*

Over 30,000 people participated in an online poll on cnn.com conducted in April 2012 asking "Have you ever driven with a pet on your lap?" We see that 34% of the particpants answered yes and 66% answered no.

a) What is the sample?

b) Is the variable in this study quantitative or categorical?

c) Can we conclude that 34% of all drivers have driven with a pet on their lap?

d) Explain why is not appropriate to generalize these results to all drivers, or even to all drivers who visit cnn.com. What is the problem with this method of data collection?

Experiments

Well-designed *experiments* incorporate four key principles:

In an *experiment* the researcher actively controls one or more of the explanatory variables.

In a *randomized experiment* the value of the explanatory variable for each unit is determined randomly, before the response variable is measured.

1. **Controlling.** Researchers assign treatments to cases and do their best to *control* any other differences between the groups. For example, when patients take a drug in pill form, some patients might take the pill with a sip of water while others drink a full glass. To control for the effect of water consumption, a doctor might instruct every patient to drink a 12-ounce glass of water with the pill.
2. **Randomization.** Researchers randomly assign subjects to treatment groups to account for variables that cannot be controlled. For example, some patients may be more susceptible to a disease than others due to their dietary habits. Randomly assigning patients to treatment or control groups helps even out such differences across groups.
3. **Replication.** The more cases researchers observe, the more accurately they can estimate the effect of the explanatory variable on the response. In a single study, we **replicate** by collecting a sufficiently large sample. At a minimum, we want multiple subjects per treatment group. Replication also refers to repeating an entire study to verify earlier findings. (The *replication crisis* refers to the ongoing problem in several scientific disciplines where past findings have failed to be replicated in new studies.)
4. **Blocking.** Researchers sometimes know or suspect that variables other than the treatment influence the response. They may first group individuals based on this variable into *blocks* and then randomize cases within each block to the treatment groups. For instance, in studying the effect of a drug on heart attacks, researchers might first split patients into low-risk and high-risk blocks, then randomly assign half the patients from each block to the treatment and half to the control group.

To demonstrate the differences between some of the more commonly used experimental designs, consider the following study. A running shoe company wants to know if the type of shoe impacts the speed of runners. One aspect of the shoe that they are considering is the heel-toe drop, or the difference in height between the heel of the shoe to the toe of the shoe. Minimalist shoes reduce the heel-toe drop to 0 – 4 mm, whereas a more traditional shoe has an 8 – 12 mm heel-toe drop. This company wants to know if a lower heel-toe drop will cause runners to run more quickly, so they recruit 300 marathon runners of varying ability. For the study, the runner will run 5k, and their time will be used to establish their pace.

Randomized comparative design

In a *randomized comparative design*, we take all the participants in the study and randomize them to a different treatment group.

In a *randomized comparative* design, we randomly assign cases to different treatment groups and then compare results on the response variable(s).

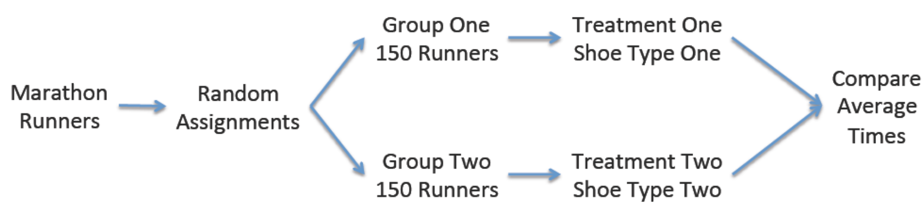

Figure 6: Randomized comparitive design.

Block design

In a *block design*, we first group participants by some logical grouping that we think might have an impact on the response in addition to the variable we are studying.

In a *block* design, cases are first split into groups (called *blocks*), and within each group the value of the explanatory variable for each unit is determined randomly.

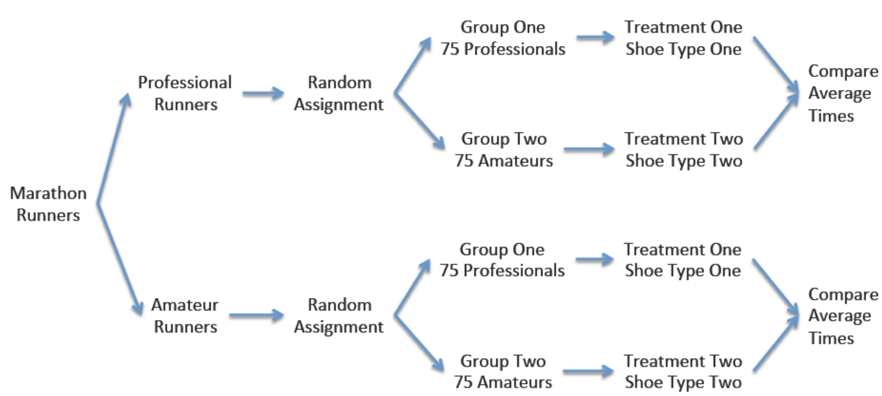

Figure 7: Block design.

Class Example 1.2.6 *Picking the blocking factor*

Why might it make sense to group the runners into professionals vs amateurs?

Matched pairs design

In a *matched pairs design*, each participant is assigned to each of two treatment settings, responses are measured for each treatment, and the differences in responses are computed for each participant. Because two measurements are obtained for each individual, the measurements are *dependent* upon each other. The two treatments often come in the form of "Before vs. After" or "Method A vs. Method B." In the "Method A vs. Method B" type of pairing, it is important to randomize the order of the treatments to avoid a confounding *order effect*.

In a *matched pairs* design, cases get both treatments (or cases are paired) and we examine differences in the response between the two treatments.

Class Example 1.2.7 *Describing the matched pairs running shoe experiment*

Thinking about the matched pairs design for the experiment about running shoes

a) What are the variables in the study? Are they categorical or quantitative?

b) What are the explanatory and response variables?

c) Explain how the treatments would be assigned.

d) Why might a matched pairs design be beneficial in this experiment?

Reducing Bias in Human Experiments

For an experiment, it is important to be able to determine if the treatment actually is making a difference, or if we are seeing results that would have happened regardless of the treatment. The purpose of a *control group* is to establish what we would expect to happen regardless of the treatment. Often times, in order to make sure that people adhere to the study guidelines, we use a *placebo* that is designed to look like the treatment.

The placebo makes it much easier to conduct *single-blinded* experiments or *double-blinded* experiments. Not all randomized experiments make use of control groups, placebos, or blinding; however, in studies involving humans, these techniques are often used to help make effective comparisons between groups.

The *control group* does not receive a treatment.

A *placebo* is a fake treatment that is designed to look like the treatment but has no therapeutic value.

In a *single-blind* study, participants do not know if they are in the treatment or control group.

In a *double-blind* study, neither the participants nor the people interacting with the participants knows if they are in the treatment or control group.

Class Example 1.2.8 *Depression and Prozac*

Researchers at a prominent Midwestern university would like to determine if Prozac is effective in reducing depression as determined by patient responses to a questionnaire. The researchers recruited 50 individuals suffering from depression. Describe a completely randomized double-blind experiment using a placebo to test this association is a causal association.

Confounding Variables

A *confounding variables* is a variable that is associated with both the explanatory variable and the response variable. Because of this dual association, a confoudning variable makes it impossible to determine whether the explantory variable truly caused the observed response.

A *confounding variable* is a third variable that is associated with both the explanatory and response variable, but not necessarily of interest in the study.

Consider a classic example: ice cream sales and drowning rates both increase during summer months. If we looked only at the association between ice cream sales and drownings, we might (absurdly) conclude that ice cream causes drowning. The confounding variable is temperature (or season) — warm weather drives both increased ice cream consumption and increased swimming, which leads to more drowning incidents.

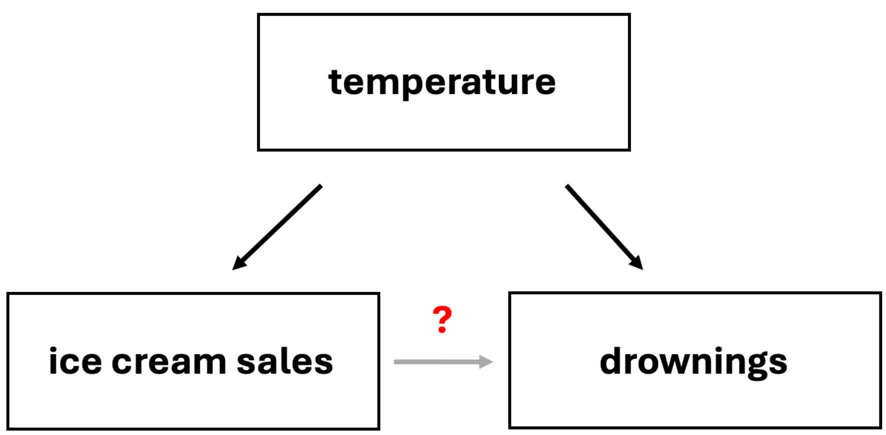

Figure 8: Confounding diagram.

Summary

In this chapter, we explored how data are collected and why the method of collection matters for the conclusions we can draw.

Key concepts:

- The *population* is the entire group of interest; a *sample* is a subset from which we collect data. A *census* collects data on every member of the population but is usually impractical.
- Good sampling methods include *simple random sampling* (*SRS*), *stratified sampling*, and *cluster sampling*. Each has advantages depending on the population's structure and practical constraints.
- *Bias* can enter through non-random sampling (selection bias), non-response, or convenience sampling.
- In an *experiment*, researchers actively assign treatments to subjects. In an *observational study*, researchers simply observe without intervening.
- *Confounding variables* are associated with both the explanatory and response variables and can create misleading associations.
- *Blinding* (single and double) and placebos help reduce bias in human experiments.

1.3 Exploring Categorical Data

> *This chapter focuses on **categorical data** — variables whose values are categories or labels rather than numbers. We begin with visualizations for a single categorical variable (bar charts, pie charts, waffle charts), then move to tools for exploring the relationship between two categorical variables (contingency tables and stacked bar charts). We close with principles for creating effective, honest data visualizations.*

Key Concepts

- Display information from a categorical variable in a table or a graph
- Use information about a categorical variable to find and report a proportion, using correct notation
- Display information about a relationship between two categorical variables in a contingency table
- Use a contingency table to find proportions

Technology for Chapter

StatLens Use the One Categorical Variable explorer for bar charts and pie charts. Use the Two Categorical Variables explorer for contingency tables and stacked bar charts.

jamovi Use Analyses>Exploration>Descriptives for bar charts for one categorical variable. Use Analyses>Frequencies>Independent Samples for two cateogrical variables.

One Categorical Variable

When we want to summarize one categorical variable, we use a *proportion* which tells us what fraction of the sample is in each group. In this course, we will use p to represent the population proportion, and $\hat{p}$ (p-hat) to represent the sample proportion. The sample proportion is given by

The *proportion* (or *relative frequency*) in a category is found by dividing the number of cases in that category by the total number of cases.

$$\hat{p} = \frac{x}{n}$$

where x is the numberof cases in a given category and n is the total number of cases.

Numerical displays

A single categorical variable is often summarized using a *frequency table* to display the number of observations that fall into each category. An alternative to the frequency table is the *relative frequency table* which displays the proportion of the sample that is in each category.

A *frequency table* is a table that displays the count for each category.

A *relative frequency table* is a table that displays the proportion of the sample in each category.

Class Example 1.3.1 *Hair Color*

The table below gives the frequency of hair color for a random sample of 55 UWL students. Find the proportions of students with each hair color and use them fill in the relative frequency column.

	Frequency	**Relative Frequency**
Brown	30	
Blonde	14	
Red	2	
Black	5	
Colorful	3	
Total	54	

Frequency table for hair color.

Graphical summaries

The standard graphical summaries for a categorical variable are the *bar chart* and the *pie chart*. A bar chart is the most common way to display the distribution of a single categorical variable. Each category gets its own bar, and the height (or length) of the bar represents either the count or the proportion of observations in that category. A pie chart divides a circle into slices, where each slice represents a category and the size of the slice corresponds to the proportion of observations in that category.

A *bar chart* includes one bar for each category, and the height of the bar is the frequency the category appears.

In a *pie chart*, the proportions correspond to the areas of sectors of a circle.

Class Example 1.3.2 *Hair Color revisited*

Sketch a bar chart and a pie chart for the hair color of random sample of 55 UWL students.

Pie charts can give a quick high-level overview, especially when a variable has only a few levels or when each level represents a simple fraction (one-half, one-quarter, etc.). However, bar charts are almost always easier to read because our eyes are better at comparing lengths than angles or areas.

Two Categorical Varaibles

Sometimes our goal is to examine whether there is a relationship between two categorical variables. To investigate these relationships, we use a *contingency table*, or *two-way table*. In a two-way table, the rows represent the groups of the first category, and the columns represent the groups of the second category. Each cell of the table contains the number of cases that are in both the row and the column category.

A *contingency table* summarizes the data for two categorical variables. The rows represent the levels of one variables, the columns represent the levels of the other.

Class Example 1.3.3 *Relationship status and class level*

169 college students were asked about their relationship status and class year (lower=freshman/sophomore, upper=junior/senior). The results are given in the table.

	Upper	**Lower**	**Total**
In a relationship	32	10	42
It's complicated	12	7	19
Single	63	45	108
Total	**107**	**62**	**169**

Contingency table for relationship status and class year.

a) What proportion of students in this sample are in a relationship?

b) What proportion of lower class students in this sample are in a relationship?

c) What proportion of upper class students in this sample are in a relationship?

d) Using $\hat{p}_U$ to represent the proportion of upper class students in a relationship and $\hat{p}_L$ to represent the proportion of lower class students in a relationship, find the difference in proportions $\hat{p}_U - \hat{p}_L$

e) What proportion of the people who are in a relationship in this sample are upper class students?

f) What proportion of the people in this sample are upper class students and in a relationship?

g) Answers c), e) and f) are all ways of representing upeer class students in a relationship. Why are they different numbers?

Congtingency tables can be displayed graphically using an extention of the bar chart. There are two common variants: a stacked bar chart and a side-by-side bar chart.

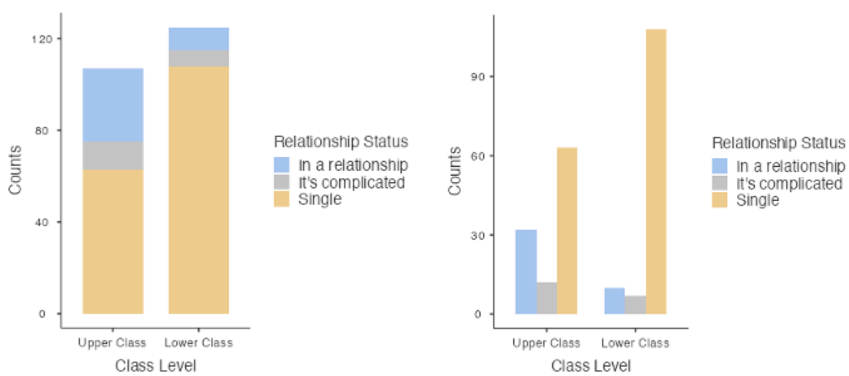

Figure 9: Bar plots for relationship status and class level.

Class Example 1.3.4 *Handedness and occupation*
In a study of handedness in occupations, 10 of 118 psychiatrists were left-handed, 26 of 148 architects were left-handed, 5 of 132 orthopedic surgeons were left-handed, and 16 of 105 lawyers were left-handed.

a) Make a two-way table of this relationship.

b) What proportion of the people in the sample are left-handed?

c) What proportion of left-handed people are architects?

d) Is your answer in part c) the same or different than the proportion of architects who are left-handed?

Effective Data Visualization

Graphs can powerfully communicate ideas directly and quickly. However, there are times when a visualization conveys a message that is inaccurate or misleading. The following are guiding principles for creating clear, honest, and effective visualizations.

- *Keep it simple*: Colors should be used purposefully and can be used to draw attention
  - Colors used only for decoration can be distracting and can confuse readers
- *Tell a story*: Include annotations (text, lines or labels that provide context beyond the raw data)
- *Order matters*: The arrangement of categories in a bar chart can help or hinder understanding
  - Alphabetical order is rarely the most informative choice
- *Make labels readable*
- *Select meaningful colors*: Default or rainbow color schemes are not always the best choice

Summary

This chapter introduced tools for exploring categorical data.

-For a single categorical variable, *bar charts* are the primary visualization, with *pie charts* reserved for simple cases.

-For two categorical variables, *contingency tables* organize counts, and stacked and side-by-side bar charts provide visual comparisons.

-Effective visualizations follow principles of simplicity, purposeful use of color, meaningful ordering, and accessibility.

1.4 Exploring Quantitative Data

> *In the previous chapter we explored categorical data. Now we turn to* ***quantitative data*** *— variables that take numerical values where arithmetic makes sense (heights, incomes, interest rates). This chapter teaches you to both see and* measure *distributions. We introduce dot plots and histograms alongside the mean and median, box plots alongside the IQR, and the standard deviation alongside histogram shapes. By the end, you will be able to describe any distribution in terms of its shape, center, spread, and unusual features.*

Key Concepts

- Understand the difference between displays of categorical and quantitative data
- Use a dotplot or histogram to describe the shape of a distribution
- Calculate the mean and the median for a set of data values, with appropriate notation
- Identify the approximate locations of the mean and the median on a dotplot or histogram
- Explain how outliers and skewness affect the values for the mean and median
- Recognize the uses and meaning of the standard deviation
- Interpret a five number summary
- Use the range, the interquartile range, and the standard deviation as measures of spread
- Describe the advantages and disadvantages of the different measures of spread
- Identify outliers in a dataset based on the IQR method
- Use a boxplot to describe data for a single quantitative variable
- Use the emperical rule to estimate the middle 68%, 95%, or 99.7% for bell-shaped distributions
- Use technology to compute summary statistics for a quantitative variable

Technology for Chapter

StatLens Use the One Quantitative Variable to visualize distributions as histograms, dot plots, or box plots, and compute means, medians, standard deviations, and quartiles for any dataset.

jamovi Use Analyses>Exploration>Descriptives to visualize distributions as histograms or box plots, and compute means, medians, standard deviations, and quartiles for any dataset.

Dot Plots and Histograms

Dot Plots

A *dot plot* is the simplest display for a single quantitative variable. Each observation is represented by a dot placed above its value on a number line. When observations share the same value (or are rounded to the same value), the dots stack vertically.

Consider the interest rates on 50 loans from a lending platform. In a dot plot, each of the 50 loans is a single dot. The red triangle marks the mean.

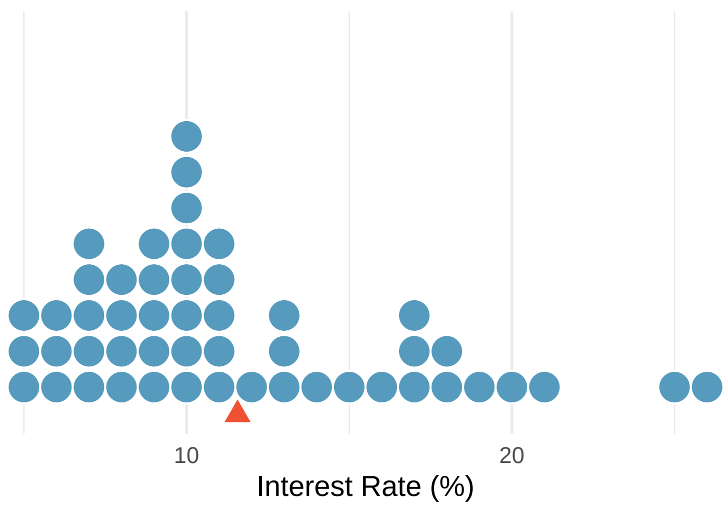

Figure 10: A dot plot of interest rates for 50 loans. Most values cluster between 5% and 15%, with a few high rates pulling the distribution to the right. The red triangle marks the mean (11.57%).

Dot plots are most useful for small datasets (up to about 50–100 observations) where you want to see every individual value. For larger datasets, histograms provide a better summary.

Histograms

A *histogram* groups quantitative data into bins (intervals) and displays the count or proportion of observations in each bin as a bar. Unlike bar charts for categorical data, histogram bars touch each other because the bins represent a continuous range of values.

To build a histogram, we:

1. Choose a bin width
2. Count how many observations fall in each bin
3. Draw bars whose heights equal the counts

Histograms are especially useful for identifying the *shape* of a distribution. One key characteristic is symmetry. When we say a distribution is *symmetric* it means that the left-side of a distribution is very similar to the right-side of the distribution. We say a distribution is *skewed left* when the tail of the distribution is longer on the left. Similarly, we say that a distribution is *skewed right* when the tail of the distribution is longer on the right.

A distribution is considered *symmetric* if we can fold the plot over a vertical center line and both sides match closely.

A distribution is considered *skewed left* if the longer tail of the distribution is to the left.

A distribution is considered *skewed right* if the longer tail of the distribution is to the right.

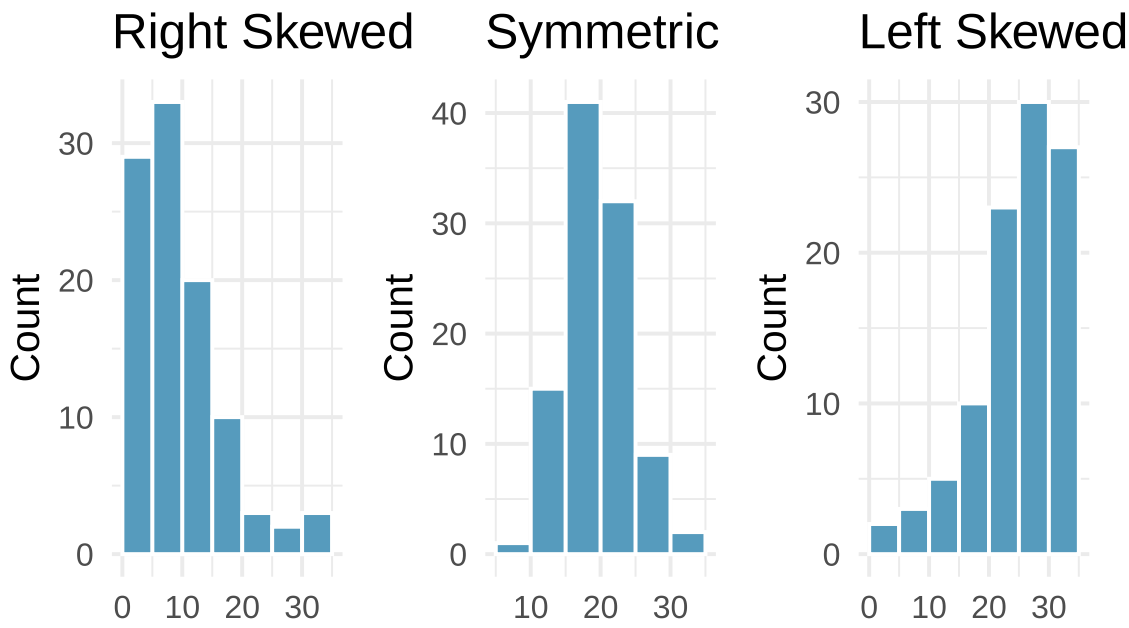

Figure 11: Three histograms depiciting the different shapes: skewed right, symmetric, and skewed left

Class Example 1.4.1 *Shape of the interest rates*

Below is the histogram for the interest rate dot plot. Describe the shape.

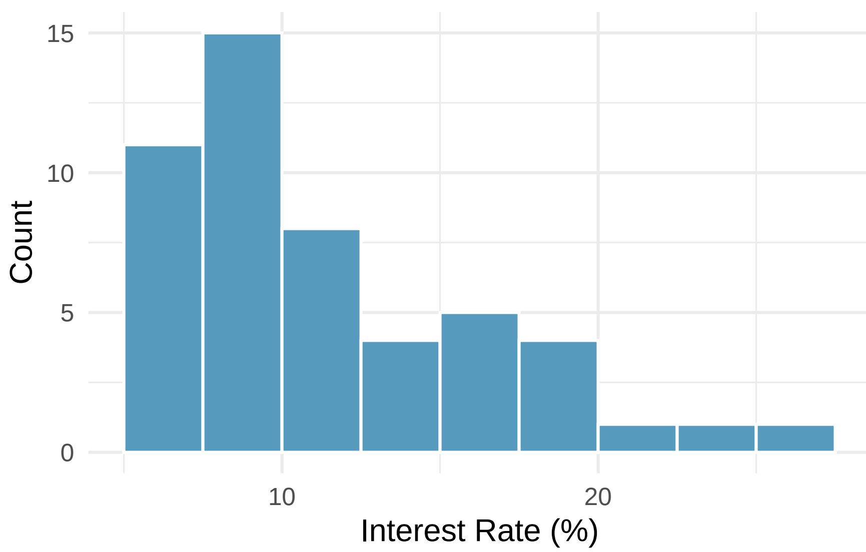

Figure 12: A histogram of interest rates with bin width 2.5 percentage points.

Measures of Center: Mean and Median

One type of numerical summaries you will see for a dataset is called measures of *center*. Measures of center provide us with a single number that best describes a quantitative variable's distribution. Essentially, they give us a guess of what a typical value would look like. There are two measures of center we will use in this class: the *mean* and *median*.

The *mean* of a data set is the sum of all its values divided by the number of observations.

The *median* of a dataset is the middle entry of an ordered list of data. If there are an even number of entries, then we take the average of the middle two values.

The mean

The *mean* of a sample tells us the average. For a sample of observations $x_1, x_2, \dots, x_n$, the mean is given by

$$\text{Mean} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$$

where n is the number of observations.

We will use μ (pronounced "mew") to represent the population mean and $\overline{x}$ (read "x-bar") to represent the sample mean. We often cannot measure μ directly because we rarely have access to every member of the population. Instead, we estimate μ using the sample mean $\bar{x}$.

Notation:
μ: population mean
$\overline{x}$: sample mean

The median

The *median*, denoted m, divides the sample into two groups of equal size. We find the median by following two steps:

1. Sort the sample values.
2. Find the middle value:
   - If n is odd, the median is the middle value in the sorted list.
   - If n is even, the median is the average (midpoint) of the middle two values in the ordered list.

The dot plot below shows both the mean and the median marked on the interest rate data. Notice that the median (9.93%) is lower than the mean (11.57%) — the few loans with very high interest rates pull the mean to the right, but the median stays anchored in the middle of the data.

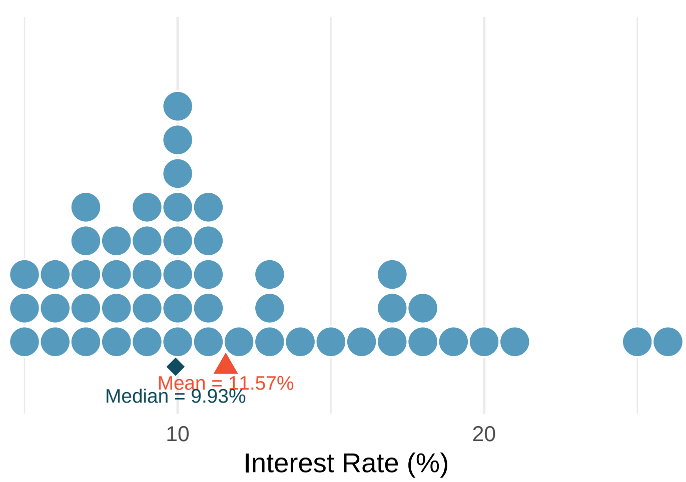

Figure 13: A dot plot of interest rates with the mean (red triangle, 11.57%) and median (blue diamond, 9.93%) marked. The mean is pulled to the right by the high-interest loans.

Class Example 1.4.2 *Ants on a sandwich*

The number of ants climbing on a piece of a peanut butter sandwich left on the ground near an anthill for a few minutes was measured 7 different times, and the results are: 43, 59, 22, 25, 36, 47, 19

a) Calculate the mean number of ants.

b) Calculate the median number of ants.

c) Suppose one of the sandwich bits was extremely appealing to the ants, and instead of 59, the number of ants for that bit was actually 159, giving the 7 values as: 43, 159, 22, 25, 36, 47, 19. Compute the mean and median for this new dataset.

d) Which statistic (mean or median) is impacted most by the outlier?

How skewness affects the mean and median

In a symmetric distribution, the mean and the median are approximately equal (mean $\approx$ median). In a right-skew distribution, the mean is pulled towards the right tail and is larger than the median (mean $>$ median). In a left-skewed distribution, the mean is pulled towards the left tail and is smaller than the median (mean $<$ median).

When to use then mean vs. the median

As we saw in the last example, the median is not impacted by extreme data values (outliers) like the mean is. We say that the median is resistant. Although the mean is a more commonly used measure of center, the median may be preferred in cases with extreme values or a high degree of skew.

If the concern is the total or aggregate, the mean is the best. If the concern is a typical observation, the median is preferred, especially when the distribution is skewed or contains extreme values.

Measures of Spread

A measure of *spread* tells us how much variability exists in the data.

Knowing the center of a distribution is important, but it tells only part of the story. Two distributions can have the same center but look very different if one is tightly clustered and the other is widely dispersed. We need measures of *spread* (also called variability or dispersion).

The range

The *range* is the difference between the maximum and minimum of the data.

The simplest measure of spread is the *range*: the difference between the maximum and minimum values.

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

While easy to compute, the range has a major weakness: it depends entirely on the two most extreme observations. A single extreme value can dramatically increase the range, giving a misleading impression of the overall variability. For this reason, we usually prefer the IQR or standard deviation.

The interquartile range (IQR)

The *IQR* (Interquartile range) is the difference between the 1st and 3rd quartiles of the data.

The *interquartile range* measures the spread of the middle 50% of the data. To find the IQR, we need to first estimate Q_1 and Q_3, the first and third quartiles. The first quartile, Q_1, is the value which 25% of the data falls below; and the third quartile, Q_3, is the value which 75% of the data falls below. To find Q_1, we need to take the median of all the values below the median. To find Q_3, we need to take the median of all the values above the median. The IQR is given by

$$\text{IQR} = Q_3 - Q_1.$$

The five-number summary

A way to combine the information from the range, the IQR, and a measure of center is through the *five-number summary*. The five-number summary consists of the minimum, Q_1, the median, Q_3, and the maximum of the data set.

Class Example 1.4.3 *Ants revisited*

In the previous example, we found that for the 7 different times, the results were 19, 22, 25, 36, 43, 47, 59 (sorted in ascending order)

a) Find the five number summary for this data.

b) Find the IQR for this data

c) Find the range for this data

d) Find the range and IQR if the value 59 was actually 159?

e) Which statistic (range or IQR) is impacted most by the extreme value?

Variance and standard deviation

The **standard deviation** is the most commonly used measure of spread. The standard deviation gives an estimate for the average distance each observation is from the mean.

We start with the concept of a **deviation**: the distance of an observation from the mean.

$$\text{deviation of } x_i = x_i - \bar{x}$$

Some deviations are positive (observations above the mean) and some are negative (observations below the mean). If we simply averaged the deviations, the positives and negatives would cancel out, giving us zero. To avoid this, we *square* each deviation before averaging. The average of the squared deviations, using $n-1$ in the denominator is the *variance* and is given by

$$\text{Variance} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n-1}$$

The *standard deviation* of a variable is a rough estimate of the typical distance of a data value from the mean.

A *deviation* is the difference between between an observation and the mean

The *variance* is the average of the squared deviations.

The *standard deviation* is the square root of the variance:

$$\text{Standard Deviation} = \sqrt{s^2} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

In this course, we will use σ^2 (pronounced "sigma squared") to represent the population variance and s^2 to represent the sample variance. We will use σ (pronounced "sigma") to represent the population standard devation and s to represent the sample standard devation. When the standard deviation is large, it means there is a lot of variability amongst the observations. When the standard deviation is small, it means there is a lot of consistency amongst the observations.

Notation:
σ: population standard deviation
s: sample standard deviation

Looking back at the interest rate data with shaded bands marking one and two standard deviations from the mean. The standard deviation of 5.05 percentage points gives us a sense of how spread out the data are around the mean of 11.57%.

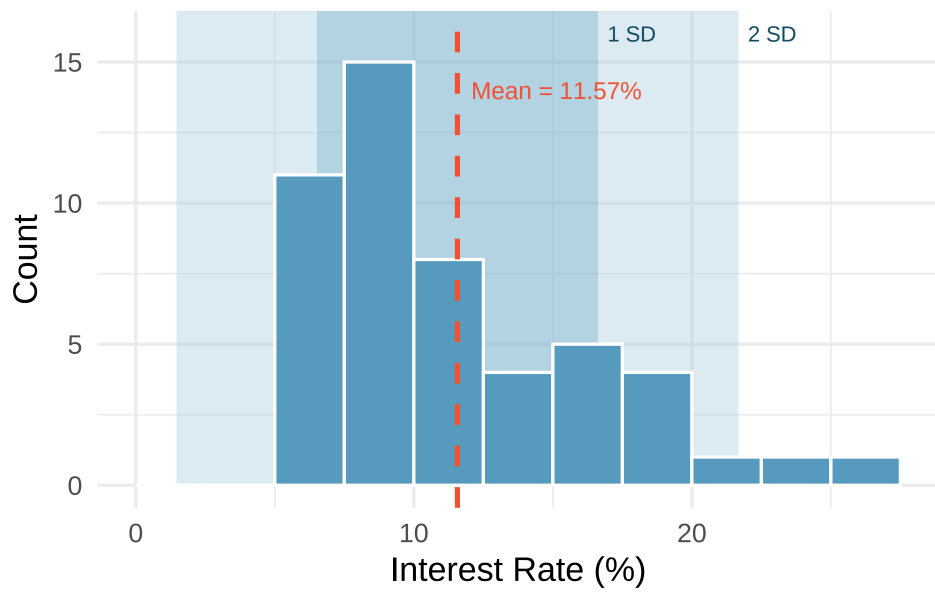

Figure 14: A histogram of interest rates with shaded regions showing one standard deviation (dark) and two standard deviations (light) from the mean.

The standard deviation tells us that interest rates typically deviate about 5 percentage points from the mean (11.57%).

Class Example 1.4.4 *Ants revisited*

a) Find the standard deviation of the ants on a sandwich data.

b) Find the standard deviation of the data if the value 59 was actually 159.

Outliers and Robust Statistics

Identifying outliers

An *outlier* is an observation that appears extreme relative to the rest of the data. A common rule for identifying outliers is the **1.5 IQR rule**: A data value, x_0 is considered to be an outlier if

$$x_0 < Q_1 - 1.5(IQR) \quad \text{or} \quad x_0 > Q_3 + 1.5(IQR).$$

An *outlier* is a data point that is unusual given the rest of the data.

Examining data for outliers serves several purposes:

- Identifying strong skew in the distribution
- Identifying possible data collection or data entry errors
- Providing insight into interesting or unusual properties of the data

Robust statistics

We call a statistic **robust** (or **resistant**) if extreme observations have little effect on its value.

We say that a statistic is *robust*, or *resistant*, if its is relatively unaffected by extreme values.

Class Example 1.4.5 *Resistant statistics*

Look back at your answers to the questions about ants on a sandwich.

a) Which measures of center (mean, median) are resistant?

b) Which measures of spread (IQR, range, standard deviation) are resistant?

Robust vs. non-robust statistics.

Robust (resistant to outliers)	**Not robust (sensitive to outliers)**
Median	Mean
IQR	Standard deviation
	Range

When a distribution is skewed or contains outliers, robust statistics (median and IQR) often give a more representative summary of the "typical" observation. When a distribution is roughly symmetric with no extreme outliers, the mean and standard deviation work well and carry additional mathematical advantages.

Box plots

So far in this class, we have used histograms and dot plots to represent data graphically. There is another very commonly used graphical summary called the *boxplot*. The boxplot gives us a graphical display of the *five-number summary* which consists of the minimum,Q_1, median,Q_3, and the maximum of a dataset.

The *boxplot* is a graphical display of the *five-number summary* which consists of the minimum,Q_1, median,Q_3, and the maximum of a dataset.

Constructing a box plot

The components of a box plot are:

1. *The box*: Spans from Q_1 (the 25th percentile) to Q_3 (the 75th percentile). This box contains the middle 50% of the data. The length of the box is the *interquartile range* (IQR = $Q_3 - Q_1$).
2. *The median line*: A line inside the box marks the median (the 50th percentile), which divides the data in half.
3. *The whiskers*: Lines extending from the box toward the smallest and largest observations that are *not* outliers. Specifically, the whiskers extend to the most extreme data points within $1.5 \times$ IQR of the box.
4. *Outlier points*: Any observations beyond $1.5 \times$ IQR from the edges of the box are plotted individually as dots. These are potential outliers.

The following gives a general sketch of a boxplot:

A box plot of interest rates shows: the left edge of the box at about 8%, the median line at about 10%, and the right edge of the box at about 14%. Two dots appear beyond the right whisker at about 25% and 26%.

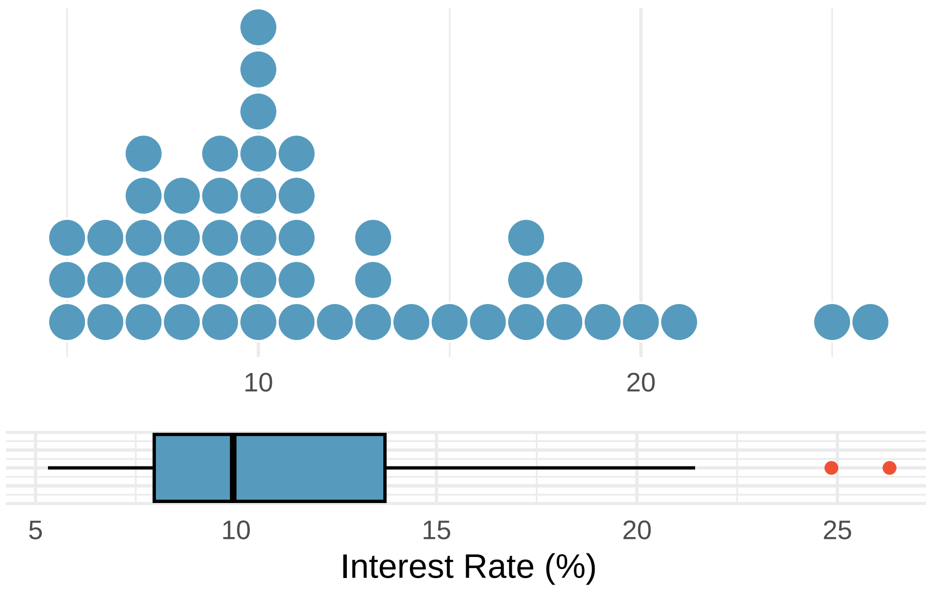

Figure 15: A dot plot and box plot of the same interest rate data, shown one above the other for comparison. The box plot summarizes the distribution with five numbers plus outliers.

Class Example 1.4.6 *Population of U.S. states*

The following data represent the populations of 12 of the 50 U.S. states, in millions of people, in ascending order.

0.506	0.830	1.315	1.813	2.333	2.953
3.524	4.198	5.504	6.207	8.685	22.472

a) Identify the five-number summary for the state populations.

b) Draw a boxplot for the data. If there are any outliers, provide a justification as to why they are outliers and identify them on your boxplot.

Class Example 1.4.7 *Gross state product*

The figure below gives a boxplot of the gestation (in days) for 54 mammal species.

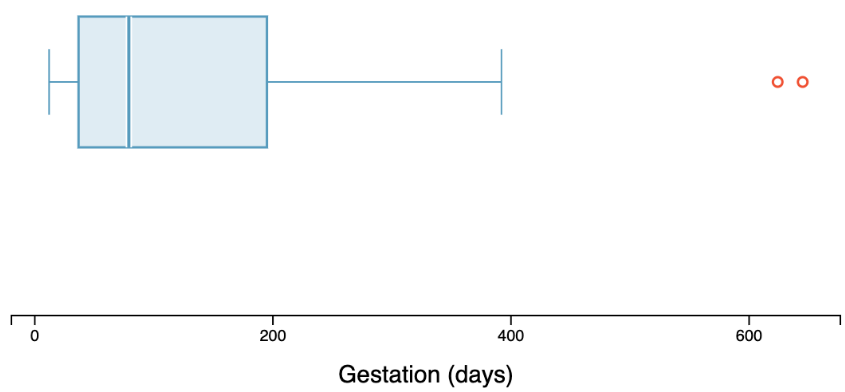

Figure 16: Box plot of gestation (in days).

a) Are there any outliers? If so, how many and estimate the values of the outliers.

b) Estimate the median, IQR and range of the data.

c) How would you describe the shape of this distribution?

d) Do you expect the mean to be less than, approximately equal to, or greater than the median? Explain.

Describing a distribution

A complete description of a distribution should address three things: *shape*, *center*, and *spread*.

A good description of a distribution should always:

1. Name the *shape*: symmetric or skewed (and direction)
2. Give the *center*: report the mean and/or median with units
3. Give the *spread*: report the standard deviation and/or IQR with units
4. Note any *unusual features*: outliers, gaps, clusters
5. Use *context*: refer to the actual variable and its units, not just abstract numbers

A way to describe a distribution is as follows: “The distribution of commute times is right skewed, with a median of 22 minutes and an IQR of 15 minutes. A few commuters have commute times exceeding 90 minutes.”

The Empirical Rule (68-95-99.7 rule)

For distributions that are approximately *bell-shaped*, the standard deviation provides a useful ruler. Statisticians call this particular bell shape the *normal distribution* — it comes up so often in statistics that it has its own name. We’ll study it formally later, but for now, just know that “bell-shaped” and “normal” mean the same thing.

The empirical rule.

For a roughly bell-shaped distribution:

- About **68%** of the data fall within 1 standard deviation of the mean:

$$(\bar{x} - s,\ \bar{x} + s)$$

- About **95%** of the data fall within 2 standard deviations of the mean:

$$(\bar{x} - 2s,\ \bar{x} + 2s)$$

- About **99.7%** of the data fall within 3 standard deviations of the mean:

$$(\bar{x} - 3s,\ \bar{x} + 3s)$$

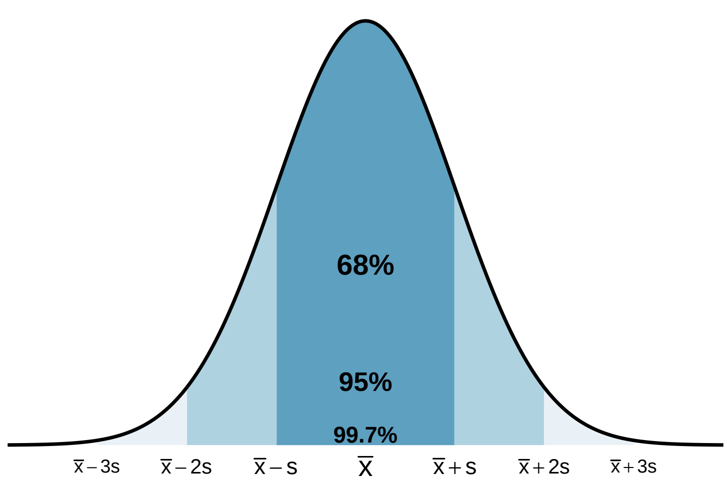

Figure 17: A bell-shaped distribution with shaded regions showing 68%, 95%, and 99.7% of the data within 1, 2, and 3 standard deviations of the mean, respectively.

Class Example 1.4.8 *SAT scores*

The SAT scores (standardized test scores) are approximately bell-shaped with a mean of 1150 and a standard deviation of 150. Use the empirical rule to estimate the range of scores for the middle 95% of students.

Summary

This chapter introduced the tools for exploring *quantitative* data — both visual and numerical.

- *Dot plots* show individual values for small datasets, *histograms* reveal the shape and density of a distribution, and *box plots* give a compact *five-number summary*.
- Distribution shape is described using symmetry (*left skewed*, *symmetric*, *right skewed*).
- The *mean* and *median* measure center, with the mean being the balancing point and the median being the middle value.
- The *standard deviation* and *IQR* measure spread, with the standard deviation roughly representing the typical deviation from the mean and the IQR representing the range of the middle 50%.
- The *median* and *IQR* are *robust* to outliers, while the mean and standard deviation are sensitive to extreme values.
- The *empirical rule* connects the standard deviation to the distribution shape for bell-shaped data: about 68%, 95%, and 99.7% of observations fall within 1, 2, and 3 standard deviations of the mean.

1.5 Relationships Between Variables

*The preceding chapters focused on exploring one variable at a time. But the most interesting statistical questions involve **relationships between variables**. Does higher education lead to higher income? Is a drug more effective for one group than another? This chapter introduces tools for exploring relationships: side-by-side box plots and faceted histograms for comparing a quantitative variable across groups, and scatterplots for examining the association between two quantitative variables. We close with the critical distinction between association and causation.*

Key Concepts

- Use a side-by-side graph to visualize a relationship between quantitative and categorical variables
- Examine a relationship between quantitative and categorical variables using comparative summary statistics

Technology for Chapter

StatLens Use the Quantitative by Group to compare distributions across groups with side-by-side box plots, histograms, and density overlays. Use the Two Quantitative Variables for scatterplots of two quantitative variables.

jamovi Use Analyses>Exploration>Descriptives, using the "Split by" input, to visualize distributions as histograms or box plots, and compute means, medians, standard deviations, and quartiles for any dataset. Use Analyses>Exploration>Scatterplot for scatterplots of two quantitative variables.

Comparing Groups

Some of the most interesting investigations involve comparing a quantitative variable across groups defined by a categorical variable. Sometimes when investigating the relationship between categorical and quantitative variables, it is helpful to plot them *side-by-side*.

We use *side-by-side* plots to visualize the relationship between a quantitative variable and a categorical variable.

Class Example 1.5.1 *Average household incomes broken down by U.S. geographical region*

The figure below shows a side-by-side boxplot of average U.S. household income broken down by geographical region of the U.S.

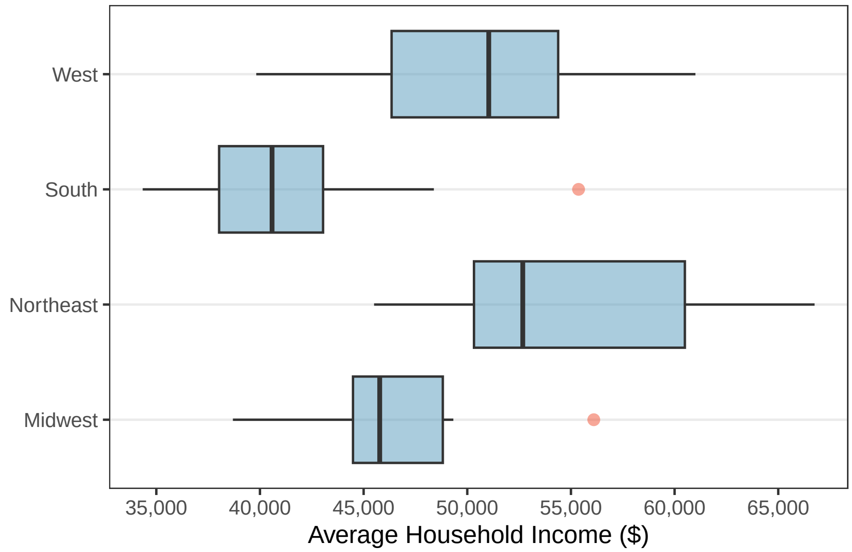

Figure 18: A side-by-side box plot of average household income split by geographic region in the US.

a) Use these side-by-side boxplots to discuss how the average U.S. household income differs by region.

b) Are there any outliers? If so, how many and estimate the values of the outliers.

c) For which region(s) would it be more appropriate to use the median instead of the mean to describe the average U.S. household income? Justify your answer.

Scatterplots

A *scatterplot* displays the relationship between two quantitative variables. Each observation is represented as a point, with one variable on the horizontal axis and the other on the vertical axis. When looking at a scatterplot, we should consider four different aspects of the plot:

A *scatterplot* is a graph of the relationship between two quantitative variables. If there are explanatory and response variables, we put the explanatory varible on the horizontal (x) axis and the response on the vertical (y) axis.

1. *Direction*: Is the association *positive* (both variables tend to increase together) or *negative* (one increases as the other decreases)?
2. *Form*: Is the relationship approximately *linear* (following a straight-line pattern) or *nonlinear* (curved)?
3. *Strength*: Is the association *strong* (points closely follow a pattern), *moderate*, or *weak* (points are loosely scattered)?
4. *Unusual observations*: Are there any *outliers* — points that deviate markedly from the overall pattern?

Example 5.2 Describing Associations

Consider a dataset of 50 loans, where we plot total income on the horizontal axis and loan amount on the vertical axis:

Figure 19: A scatterplot of loan amount versus total income for 50 loans. There is a moderate positive association.

Describe the association seen in the scatterplot in Figure 2.

Associations vs. causation

We have explored the association between variables, but an association is not the same as causation. When two variables are associated, it mease they tend to vary together. That does not necesasrily mean one variable *causes* the other. There may be a *confounding varible* that is related to both, creating the illusion of a direct relationship.

A *confounding variable* is a third variable that is associated with both the explanatory and response variable, but not necessarily of interest in the study.

Example 5.3 GPAs and Breakfast

A study finds that students who eat breakfast tend to have higher GPAs than students who skip breakfast. Can we conclude that eating breakfast causes higher GPAs?

Summary

This chapter introduced tools for exploring relationships between variables.

- For comparing a *quantitative variable across groups* defined by a categorical variable, side-by-side box plots provide a compact comparison of centers and spreads, faceted histograms reveal full distributional shapes, and ridge plots overlay density curves for compact multi-group comparison.
- For *two quantitative variables*, scatterplots reveal the direction, form, strength, and unusual features of an association.
- We closed with the fundamental principle that *association does not imply causation* — confounding variables can create misleading associations, and establishing causation generally requires a randomized experiment.

2.1 *Sampling Variability*

So far in this course, you have learned how to collect data, visualize it, and compute summary statistics. But statistics is about more than describing the data you have — it is about using data to learn about a larger population. The central challenge is this: ***every sample is different****. If you took another sample from the same population, you would get different results. This chapter introduces the foundational ideas that make statistical inference possible: the distinction between parameters and statistics, the concept of a sampling distribution, and the standard error as a measure of how much sample statistics vary. These ideas are the bridge from describing data to drawing conclusions about the world.*

Key Concepts

- Distinguish between a population parameter and a sample statistic, recognizing that a parameter is fixed while a statistic varies from sample to sample
- Compute a point estimate for a parameter using an appropriate statistic from a sample
- Recognize that a sampling distribution shows how sample statistics tend to vary
- Recognize that statistics from random samples tend to be centered at the population parameter
- Estimate the standard error of a statistic from its sampling distribution
- Explain how sample size affects a sampling distribution

Parameters vs. Statistics

In Unit 1, *statistical inference* was defined as the process of using a *statistic* calculated from a sample to estimate a *parameter*, or unknown value, from the population. The reason this cycle works is that when we gather data properly (i.e. use a random sampling technique), the statistics behave in a predictable way. Sample statistics are rarely, if ever, the exact population parameter, but they are usually close. If we only have one sample and we don't know the value of the population parameter, the sample statistic is our best estimate of the true population parameter (called a *point estimate* since it's a single number.)

Statistical inference is using the information in a sample to draw conclusions about the entire population.

A *parameter* is a number that describes some aspect of the population.

A *statistic* is a number that is computed from the data in a sample.

We use the sample statistic as a *point estimate* for the population

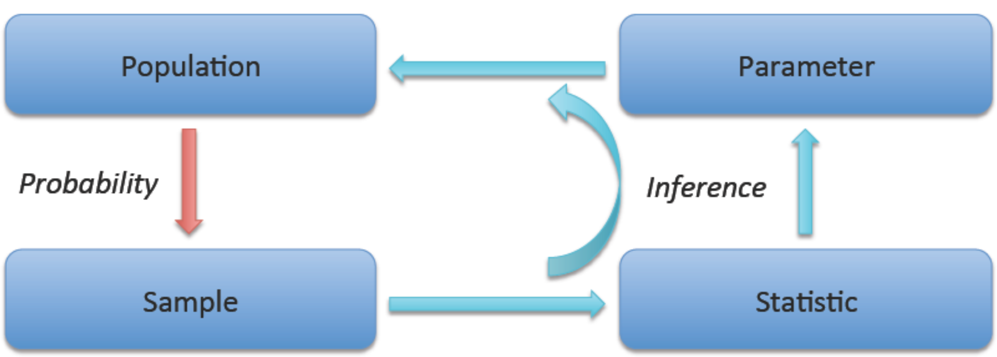

Figure 20: The cycle of statistical inference.

We use specific notation to distinguish parameters from statistics:

Quantity	**Parameter (population)**	**Statistic (sample)**
Proportion	p	$\hat{p}$
Mean	μ	$\bar{x}$
Standard deviation	σ	s
Difference in proportions	$p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$
Difference in means	$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$

Class Example 2.1.1 *Parameter vs. Statistic*

For each of the following, determine whether the quantity described is a parameter or a statistic and give the proper notation.

a) Average height of players on the 2024 Brazil World Cup team, using data from all 23 players on the roster.

b) Average daily high temperatuer in La Crosse in June, based off of randomly selecting 100 June days over the last 20 years.

c) Proportion of households in the U.S. who are renting their living accommodations using data from the 2020 Census.

d) Proportion of UWL students who live on campus, using data from 30 students.

Sampling Distributions

To understand how far a statistic might be from its parameter, we need to understand what happens when we take many samples from the same population. A sampling distribution is the distribution of sample statistics computed for different samples of the same size drawn from the same population. The sampling distribution shows us how the sample statistic varies from sample to sample and gives us a way to see how close we can expect statistics to be to the true parameter value.

The *sampling distribution* shows how a statistic will vary from sample to sample.

Class Example 2.1.2 *Illustrating sampling distributions*

Go to StatLens at https://learnlens.org/statlens/ and click on Sampling Distributions under the Simulate Tab. We will use the Right-Skew population shape. We do not have access to the population, but we wish to estimate the (then unknown) population mean.

a) What is the popuation mean (μ)?

b) Use StatLens to select a sample size of 10. What was the sample mean ($\overline{x}$) of this data set? Is it close to the popuation mean?

c) Keep taking samples of size 10, one by one. Visualize how the sample means of these samples change. Even though the sample means change from sample to sample, where do they appear to center?

d) Now generate 1000 samples at once to get a fuller picture of the sampling distribution of the sample mean. Where is the distribution centered? What range do the sample means lie within?

e) Reset the plot and now consider drawing samples of size 100 and exploring the sample mean. Begin with drawing samples one by one. Where do the sample means tend to center? Do they deviate much from this center?

f) Now generate 1000 samples at once. Where is the sampling distribution centered? What range do the sample means lie within? How does this range compare to the range from part d)?

g) What accounts for the differences in variability from part d) to part f)?

Facts about sampling distributions

For many common statistics, if the sample size is large enough and representative of the population (i.e. a random sample), the sampling distribution has three important features:

1. *Center*: The sampling distribution is centered at the population parameter. On average, the sample statistic equals the parameter.
2. *Spread*: The sampling distribution has a measureable spread that decreases as the sample size increases. Larger samples give more precise estimates.
3. *Shape*: For many statistics (e.g. $\overline{x}$ and $\hat{p}$), the sampling distributionis approximately bell-shaped (normal) when the sample size is large enough.

Sampling distributions tell us useful information about which sample statistics are likely to occur. Sample statistics that fall near the center of the sampling distribution are more likely to occur than those that fall far from the center.

The *sampling distribution* is fundamentally different from the *data distribution*:

- The data distribution shows how individual observations vary.
- The sampling distribution shows how a statistic (like the mean or proportion) varies from sample to sample.

Variability of sample statistics

When gathering data, there is a tradeoff between how large a sample is and the accuracy of the statistic calculated from the sample. This happens because with a larger sample size, we get a more complete picture of what the population looks like. So, as the sample size increases, we are more likely to see sample statistics that are closer to the true value of the population parameter (i.e. the variability of the sample statistics tends to decrease). We saw this in the last example where we explored the sampling distribution for the mean in StatLens.

As the sample size *increases* the variability of a statistic will *decrease*.

Two factors control how much a sample statistic bounces around from sample to sample:

1. *Sample size* (n): Larger samples produce statistics that are closer to the parameter. This is intuitive — a sample of 1,000 tells you more about the population than a sample of 10.

2. *Population variability*: If the population is very homogeneous (everyone is similar), then any sample will look like any other sample, and the statistic won't vary much. If the population is very heterogeneous (lots of individual differences), then different samples can look quite different, and the statistic will vary more.

Class Example 2.1.3 *Average pH levels in Florida lakes*

The figure below shows two sampling distributions for the average pH level in Florida lakes for two different sample sizes.

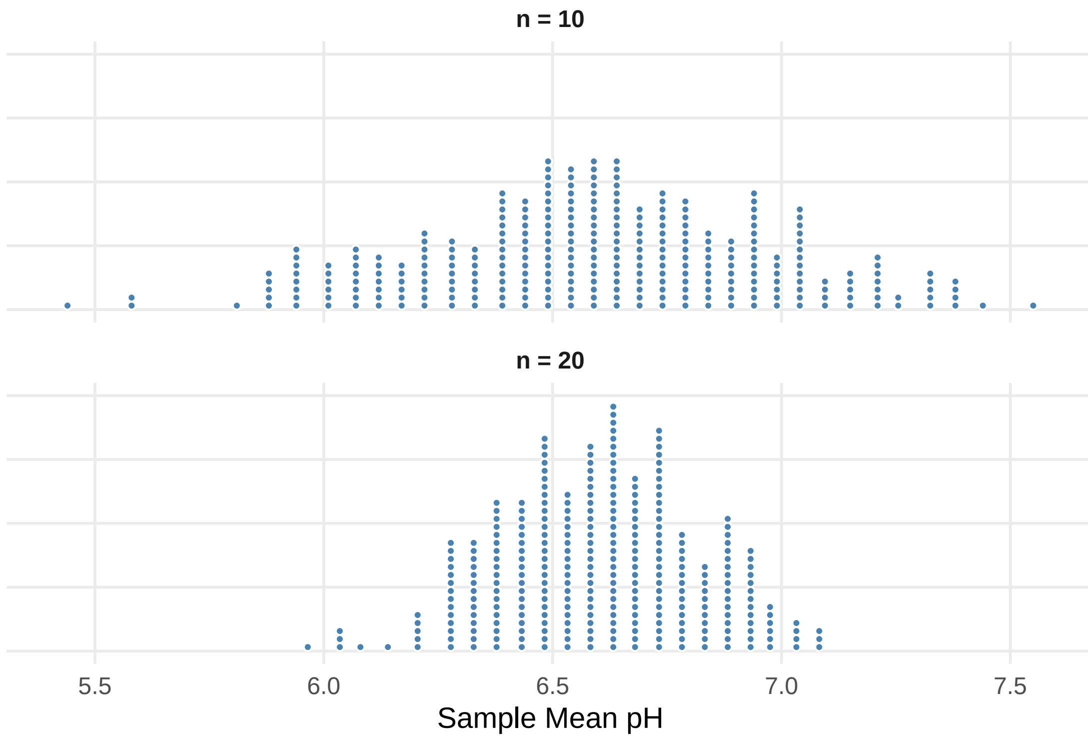

Figure 21: Two sampling distributions for average pH levels in Florida lakes.

a) The difference between these two dotplots is that one comes from samples where $n = 10$ and the other comes from samples where $n = 20$. Identify which dotplot corresponds to each sample size.

- $n = 10$ Top Bottom
- $n = 20$ Top Bottom

b) What does each dot in the dotplots represent?

c) For which sample size are we more likely to find an average pH greater than 7?

- Circle one: $n = 10$ $n = 20$

In a sampling distribution the *standard error* is the standard deviation of the statistic.

An *inflection point* is a point on a distribution where the up-cup becomes the down-cup.

Class Example 2.1.4 *Estimating standard errors from a dot plot*

Refer back to the Figure in the previous example. Use the method above to estimate the standard error for each data set.

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2.2

Key Concepts

- Recognize when and why statistical tests are needed
- Specify null and alternative hypotheses based on a question of interest, defining relevant parameters
- Interpret a p-value as the probability of results as extreme as the observed results happening by random chance, if the null hypothesis is true
- Estimate a p-value from a randomization distribution
- Comparatively quantify strength of evidence using p-values
- Distinguish between one-tailed and two-tailed tests in estimating p-values
- Make a formal decision in a hypothesis test by comparing a p-value to a given significance level
- State the conclusion to a hypothesis test in context
- Interpret Type I and Type II errors in hypothesis tests
- Recognize a significance level as measuring the tolerable chance of making a Type I error
- Recognize that statistical descernibility is not always the same as practical significance

Figure 22: Boxes for the cartoon marketing cereal study.

a)

b)

c)

A *hypothesis test*, or *statistical test* uses data from a sample to assess a claim about a population.

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Usually, a *null hypothesis*, notated H_0, is a claim that there really is "no effect" or "no difference", or represents the status quo.

An *alternative hypothesis*, notated H_A, is the claim for which we seek discernable evidence, and is established by observing evidence that contradicts the null hypothesis and supports the alternative hypothesis.

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Class Example 2.2.1 *Defining statistical hypotheses*

For each of the following scenarios, state the null and alternative hypotheses with the relevant parameter(s).

a) We are testing to see if the average income is different in eastern states than western states.

b) We are testing to see if the average income is greater in eastern stats than western states.

c) We are testing to see if a particular coin is weighted (i.e., not fair).

d) We are testing to see if the average rent in La Crosse is less than \$850.

e) We are testing to see if children have a preference for the cereal box with cartoons (above).

Class Example 2.2.2 *Opportunity cost*

Using the opportunity cost case study, answer the following questions:

a) Is this an observational study or an experiment? How does the thype of study impact what can be inferred from the results?

b) State the null and alternative hypotheses for this study.

Class Example 2.2.3 *Opportunity cost, revisited*

Compute the difference in the sample proportions of students who choose not to buy the video.

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A *randomization distribution* approximates a sampling distribution from a population where the null hypothesis is true.

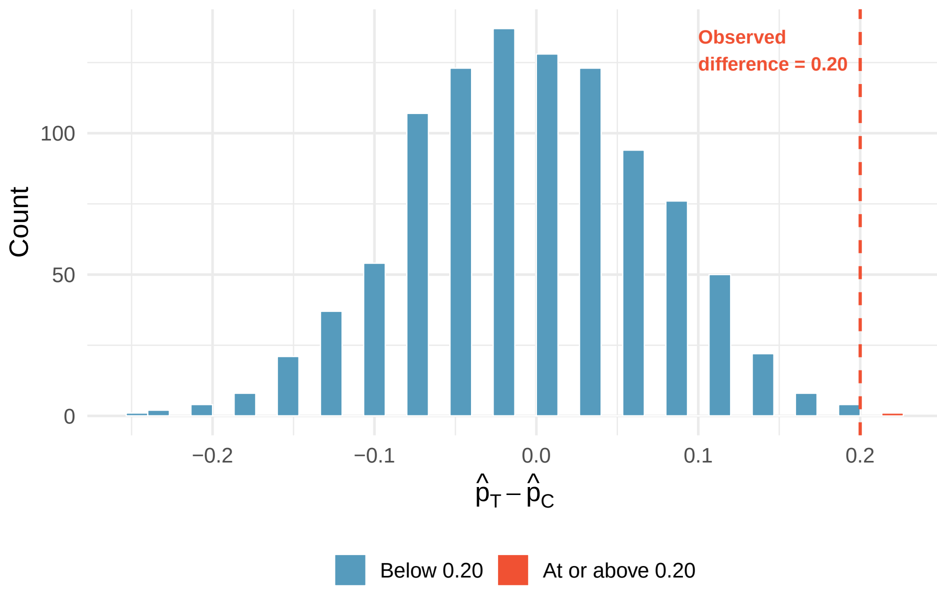

Figure 23: A histogram of 1,000 simulated differences produced under the null hypothesis. Differences at least as large as the observed 0.20 are highlighted in red.

The p-*value* of the sample data in a statistical test is the probability of obtaining a sample statistic as extreme as (or more extreme than) the observed sample when the null hypothesis is *true*.

In order to correctly estimate the p-value from a randomization distribution, we need to keep our alternative hypothesis in mind!

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Class Example 2.2.4 *Estimating a p-value from a randomization distribution*

The figure below shows a randomization distribution for testing $H_0 : p = 0.3$ vs $H_A : p < 0.3$. In each case, use the distribution to decide which value is closer to the p-value for the observed sample proportion. There are 1000 randomizations given in the figure below.

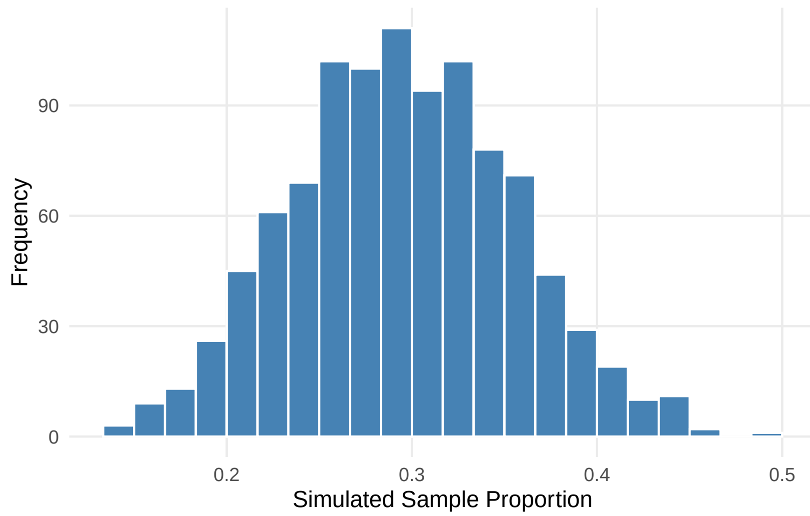

Figure 24: Randomization distribution with 1000 randomizations

a) Why is the distribution centered at 0.3?

b) The p-value for $\hat{p} = 0.15$ is closest to: 0.04 or 0.40?

c) Which proportion, $\hat{p} = 0.20$ or $\hat{p} = 0.10$, provides the strongest evidence to suggest that H_0 is not true? Explain.

Class Example 2.2.5 *Estimating a p-value from a randomization distribution…continued*

The figure below shows a randomization distribution for testing $H_0 : \mu_1 - \mu_2 = 0$ vs $H_A : \mu_1 - \mu_2 \neq 0$. The statistic used for each sample is $D = \overline{x}_1 - \overline{x}_2$. In each case, use the distribution to decide which value is closer to the p-value for the observed sample difference in averages. There are 1000 randomizations given in the figure below.

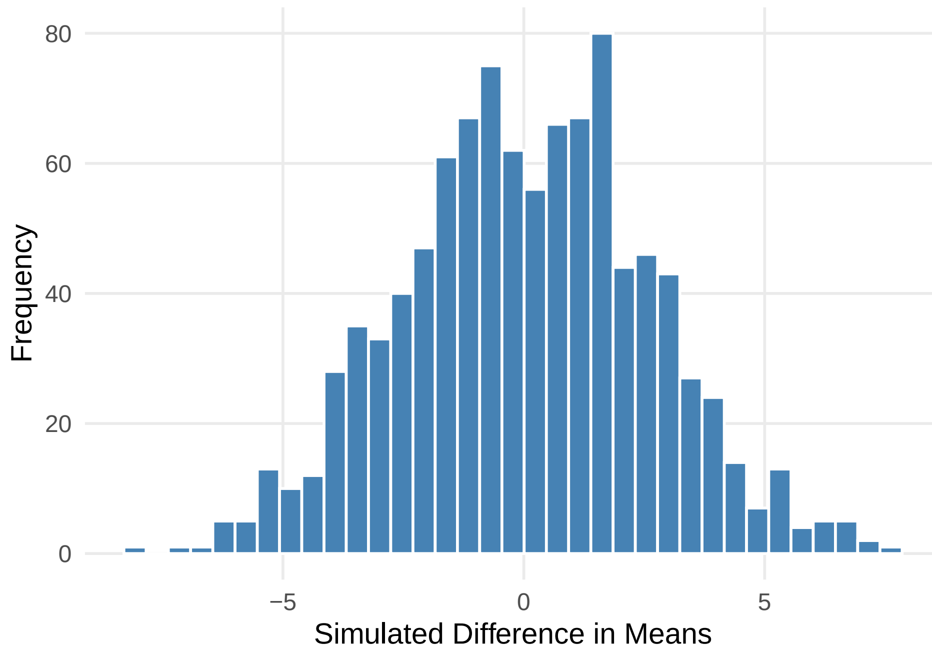

Figure 25: Randomization distribution with 1000 randomizations

a) The p-value for $D = \overline{x}_1 - \overline{x}_2 = -2.9$ is closest to: 0.01 or 0.25?

b) The p-value for $D = \overline{x}_1 - \overline{x}_2 = 1.2$ is closest to: 0.30 or 0.60?

c) Of the two values of D above, which provides strongest evidence to suggest that H_0 is not true? Explain.

Class Example 2.2.6 *Migraine Acupuncture Trial*

Can acupuncture relieve migraine pain? Allais et al. (2011) conducted a randomized controlled study where 89 individuals who suffered from migraines were randomly assigned to receive either acupuncture (treatment, $n = 43$) or placebo acupuncture (control, $n = 46$). The placebo acupuncture involved placing needles at non-acupuncture-point locations. Researchers recorded whether each patient was pain-free 24 hours after treatment. The full data are in the data set *Migrane Acupuncture Trial* on StatLens.

a) What are the null and alternative hypotheses for this study?

b) What was the sample statistic for these data?

c) What is the probability that the sample results (or those more extreme) occurred by random chance, assuming that the acupuncture had no effect?

d) How would your answer to c) change if the alternative hypothesis was two-sided?

Class Example 2.2.7 *Calorie Counts*

Is the average amount of kilocalories from a major U.S. fast food chain menu item more than 525 kilocalories? Calorie counts for 515 menu items from eight major fast food chains were collected. The full data are in the data set *Fast Food Calories* on StatLens.

a) State the null and alternative hypotheses

b) What is the sample statistic?

c) To create the randomization distribution, what do we have to assume?

d) What is the p-value for this study?

Class Example 2.2.8 *Cereal study revisited*

Recall the movtivating cereal study. We were trying to determine if children had a preference for the cereal in the box with the cartoon. In the study 52 out of 80 children chose the box with cartoons. Use StatLens to find the p-value for this scenario.

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When we make a formal decision, we have two options:

- Reject H_0
- Do not reject H_0

If we reject the null hypothesis, we say our finding is *statistically discernable*.

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We use the *decernibility level*, α, to decide the cutoff for our p-value to be discernable.

Common values for $\alpha : 0.05, 0.01, 0.10$

We reject H_0 when p-value $< \alpha$

We do not reject H_0 when p-value $\geq \alpha$

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Class Example 2.2.9 *Red wine and weight loss*

Resveratrol, an ingredient in red wine and grapes, has been shown to promote weight loss in rodents. A recent study (*Science Daily*, 2010) investigates whether the same phenomenon holds true in primates. A sample of six lemurs had their resting metabolic rate, body mass gain, food intake, and locomotor activity measured for one week prior to resveratrol supplementation and then the four indicates were measured again after treatment. For each of the following, use the p-value to make the appropriate conclusion (reject or do not reject) for the hypothesis tests. Use $\alpha = 0.05$ for each of your decisions.

a) In a test to see if mean resting metabolic rate is higher after treatment (p-value of 0.013).

b) In a test to see if mean body mass gain is lower after treatment (p-value of 0.007).

c) In a test to see if mean locomotor activity is affected by treatment (p-value of 0.980).

Class Example 2.2.10 *Red wine and weight loss revisted*

For each of the following, provide an interpretation of the conclusion in the context of the original study.

a) In a test to see if mean resting metabolic rate is higher after treatment (p-value of 0.013).

b) In a test to see if mean body mass gain is lower after treatment (p-value of 0.007).

c) In a test to see if mean locomotor activity is affected by treatment (p-value of 0.980).

Class Example 2.2.11 *Migraine Acupuncture Trial revisited*

Can acupuncture relieve migraine pain? Using your decision from the Acupunture Trial example, interpret your results in the context of the problem.

Class Example 2.2.12 *Calorie count revisited*

Is the average amount of kilocalories from a major U.S. fast food chain menu item more than 525 kilocalories? Using your decision from the Calorie Count example, interpret your results in the context of the problem.

Class Example 2.2.13 *Cereal study revisited*

Recall the movtivating cereal study. Interpret the results of the study in the context of the problem.

A *Type I error* occurs when we reject H_0, but it is true (e.g., convict an innocent person).

A *Type II error* occurs when we do not reject H_0, but it is false (e.g., let a guilty person go free).

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Class Example 2.2.14 *Red wine and weight loss revisted*

For each of the following, identify which kind of error could have occured (Type I or Type II) using $\alpha = 0.05$.

a) In a test to see if mean resting metabolic rate is higher after treatment (p-value of 0.013).

b) In a test to see if mean body mass gain is lower after treatment (p-value of 0.007).

c) In a test to see if mean locomotor activity is affected by treatment (p-value of 0.980).

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The descernability level, α, is the probaiblity we make a Type I error.

The *power* of a test tells us how likely we are to reject H_0 when we should.

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The *power* of a test tells us how likely we are to reject H_0 when we should

Class Example 2.2.15 *Errors in a drug study*

A pharmaceutical company is researching a new anti-anxiety drug. Marketing a new drug is costly, and the company only makes a profit when the drug is effective for over 80% of patients. A random sample of 60 participants are placed on the drug and the number of participants that respond favorably to the drug will be recorded.

a) Give the null and alternative hypothesis that you would use for determining if the drug should be marketed.

b) Describe in words what a Type I error would mean in the context of this study.

c) Describe in words what a Type II error would mean in the context of this study.

d) Suppose the significance level is set at a = .05, but the researchers find that this produces a Type II error rate of 0.4 and this is too high for their liking. List two ways they could reduce the Type II error rate.

e) Given a Type II error rate of 0.4, what is the power of the test?

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2.3

Key Concepts

- Construct a confidence interval for a parameter based on a point estimate and a margin of error
- Use a confidence interval to recognize plausible values of the population parameter
- Construct a confidence interval for a parameter given a point estimate and an estimate of the standard error
- Interpret (in context) what a confidence interval says about a population parameter
- Recognize that a bootstrap distribution tends to be centered at the value of the original statistic
- Use technology to create a bootstrap distribution
- Estimate the standard error of a statistic from a bootstrap distribution
- Construct a 95% confidence interval for a parameter based on a sample statistic and the standard error from a bootstrap distribution
- Construct a confidence interval based on the percentiles of a bootstrap distribution
- Explain how the width of an interval is affected by the desired level of confidence and the sample size
- Recognize when it is appropriate to construct a bootstrap confidence interval using percentiles or the standard error

An *interval estimate* gives a range of plausible values for a population parameter, commonly formed using *point estimate*$\pm$*margin of error*.

Margin of error is a number that reflects the precision of the sample statistic as a point estimate for the parameter.

Class Example 2.3.1 *Adopting a child in the US*

A survey of 1000 American adults conducted in January 2013 stated that "44% say it's too hard to adopt a child in the US." The survey goes on to say that "The margin of error is ± 3 percentage points with a 95% level of confidence."

a) What is the relevant sample statistic? Give appropriate notation and the value of the statistic.

b) What population parameter are we estimating with this sample statistic?

c) Use the margin of error to give a confidence interval for the estimate. Is 0.42 a plausible value of the population proportion? Is 0.50 a plausible value?

d) If we took a sample of 100 instead, do you expect the margin of error to increase or decrease?

A *confidence interval* is a range of plausible values for the population parameter.

The success rate (proportion of all samples whose intervals contain the parameter) is known as the *confidence level*.

Class Example 2.3.2 *Interpreting confidence level*

For each of the following indicate the number of confidence intervals that should capture the true population parameter.

a) We repeat our sampling 30 times and calculate 95% confidence intervals each time.

b) We repeat our sampling 100 times and calculate 99% confidence intervals each time.

A *95% confidence interval* is constructed using a procedure that, when applied repeatedly to many different sample, caputers the true parameter in approximately 95% of all intervals produced

Using the standard error, the *95% confidence interval* for a sampling distribution that is relatively symmetric and bell shaped is formed using *Statistic* $\pm 2 \cdot SE$.

Class Example 2.3.3 *Constructing confidence intervals*

For each of the following, use the information to construct a 95% confidence interval and give notation for the quantity being estimated.

a) $\overline{x} = 27$ with standard error 3.2.

Interval: Notation (circle one): p, μ, $p_1 - p_2$, $\mu_1 - \mu_2$

b) $\hat{p}_1 - \hat{p}_2 = 0.05$ with margin of error for 95% confidence of 0.02.

Interval: Notation (circle one): p, μ, $p_1 - p_2$, $\mu_1 - \mu_2$

For a 95% confidence interval: $MOE = 2 \cdot SE$

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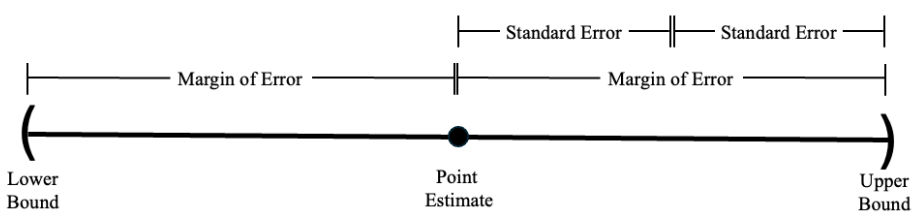

Figure 26: Diagram of the relationship between the 95% confidence interval, margin of error, and standard error.

Class Example 2.3.4 *Have you ever been arrested?*

According to a recent study of 7335 young people in the US, 30% had been arrested for a crime other than a traffic violation by the age of 23. Crimes included such things as vandalism, underage drinking, drunken driving, shoplifting, and drug possession.

a) Is the 30%, that the study reports, a parameter or statistic?

b) The margin of error is 0.01 for a 95% confidence interval. Use this information to give a range of plausible values for the parameter.

c) A correctional facility network claims that the true proportion of young people who had been arrested for a crime other than a traffic violation is less than 25%. Given the margin of error in part b), if we asked *all* young people in the US if they have ever been arrested, is it likely that the actual proportion is less than 25%?

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Class Example 2.3.5 *BPA in soup cans*

To investigate BPA exposure from cans, 75 participants ate either canned soup or fresh soup for lunch for five days. On the fifth day, urinary BPA levels were measured. After a two-day break, the participants switched groups and repeated the process (matched pairs experiment). The difference in the BPA levels between the canned and fresh soup were measured for each participant, yielding a 95% confidence interval for the difference in means (canned minus fresh) of 19.6 to 25.5 micrograms/L.

a) Provide an interpretation of the confidence interval in the context of the study.

b) What is the margin of error for this confidence interval.

c) Does this inteval suggest that BPA levels differ for canned and fresh soup? Explain.

d) If the study had included 500 participants instead of 75, would you expect the confidence interval to be wider or narrower? Explain.

Class Example 2.3.6 *Biomass in tropical forests*

Using a sample of 4079 inventory plots, scientists give a 95% confidence interval of 9600 to 13600 tons for the mean amount of carbon per square kilometer in tropical forests.

a) What is wrong with this interpretation? "We believe that 95% of tropical forests will have an average of carbon per square kilometer that is between 9600 to 13600 tons."

b) Provide a correct interpretation of the confidence interval in the context of the study.

Class Example 2.3.7 *Tipping for pizza delivery*

Using a sample of 24 deliveries described in "Diary of a Pizza Girl", we find a 95% confidence interval for the mean tip given for a pizza delivery to be \$4.18 to \$5.90. Which of the following is a correct interpretation of this interval? Circle the correct interpretation(s).

a) I am 95% confident that all pizza delivery tips will be between \$4.18 and \$5.90.
b) 95% of all pizza delivery tips will be between \$4.18 and \$5.90.
c) I am 95% confident that the mean pizza delivery tip for this sample will be between \$4.18 and \$5.90.
d) I am 95% confident that the mean tip for all pizza deliveries in this area will be between \$4.18 and \$5.90.
e) I am 95% confident that the confidence interval for the mean pizza delivery tip will be between \$4.18 and \$5.90.

A *point estimate* is a single numerical value used as a "best guess" or approximation to estimate an unknown population parameter.

The *bootstrap distribution* is a collection of *bootstrap statistics*, each computed from a *bootstrap resample*. The boostrap resample is a sample of size n drawn with replacement from the original sample.

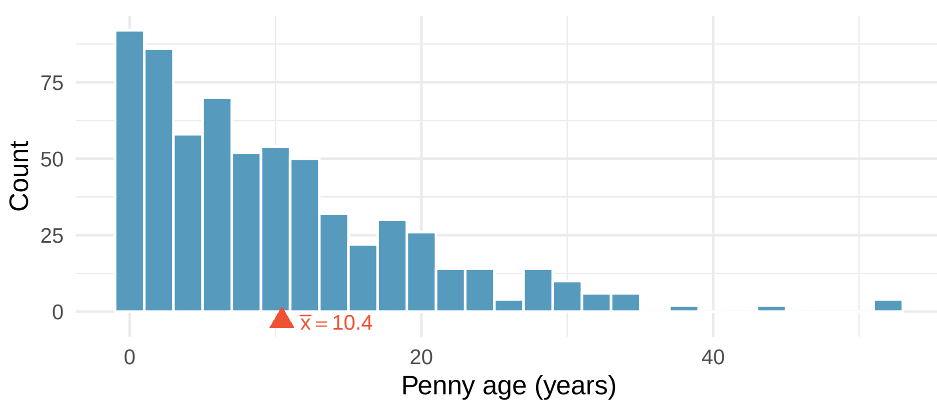

Figure 27: Distribution of ages for 648 pennies sampled from circulation. The distribution is right-skewed with a mean of about 10.4 years. The sample mean is marked with a red triangle. Open in StatLens

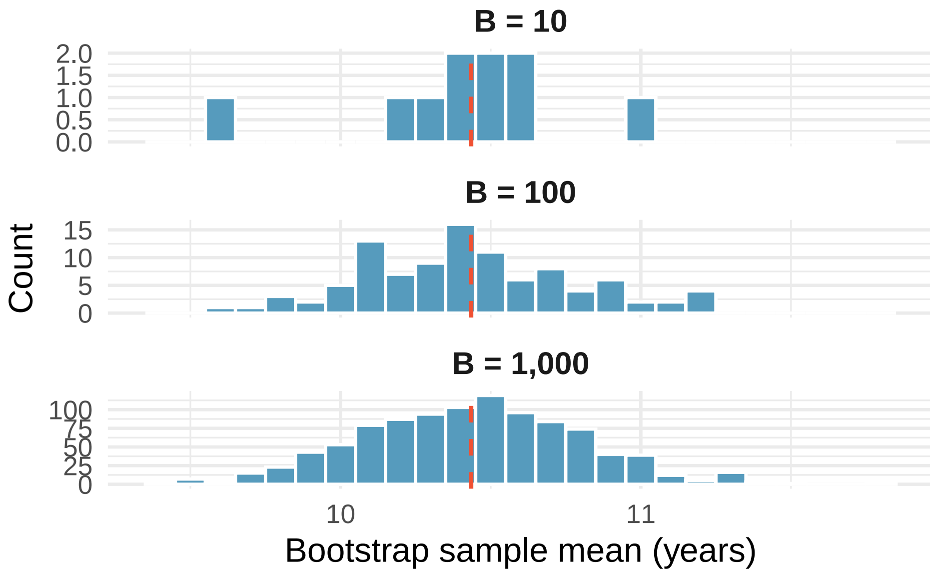

Figure 28: Building up the bootstrap distribution. With 10 resamples we see only a rough sketch; by 1,000 resamples the shape of the distribution is clear — and approximately bell-shaped (normal), even though the original data are skewed.

The *Pth percentile* is the value of a quantiative variable which is greater than P percent of the data

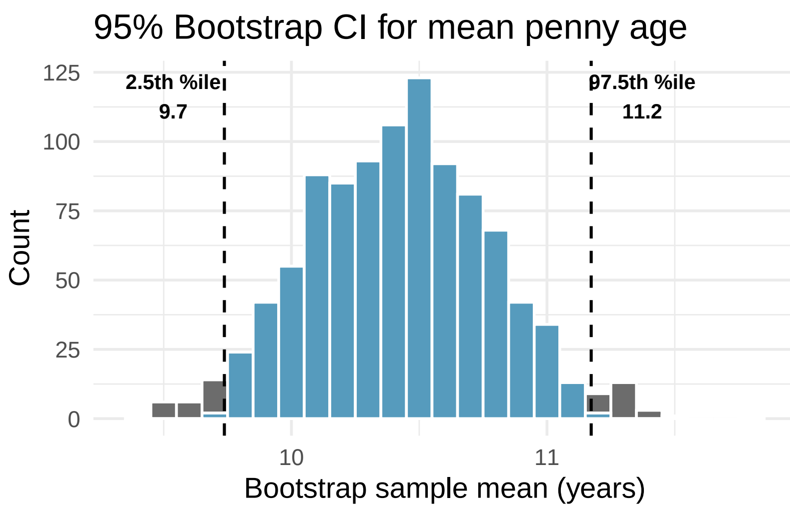

Figure 29: Bootstrap distribution of 1,000 sample means with the middle 95% shaded in blue. The 2.5th and 97.5th percentiles form the 95% confidence interval for the mean penny age.

Figure 30: Twenty 95% confidence intervals from 20 different random samples of pennies. The true population mean is shown as a vertical dashed line. Approximately 95% of intervals capture the true mean; a few (shown as red and dotted) miss it.

Class Example 2.3.8 *Can these be a bootstrap sample?*

We have a random sample of 5 exam grades: 85, 72, 79, 97, 88. For the following state whether or not it is a possible bootstrap sample. If it is not possible, provide a correction to the sample.

a) 79, 79, 97, 85, 82

b) 85, 88, 97, 72

c) 85, 85, 85, 85, 85

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Class Example 2.3.9 *Increasing confidence level*

If you increase the confidence level from 95% to 99%, what happens to the width of the confidence interval? Why?

Class Example 2.3.10 *Average penalty minutes in the NHL*

We have a random sample of the penalty minutes for $n = 26$ players from the 2018–2019 season for the NHL team the Ottawa Senators. Some percentiles from a bootstrap distribution of 1000 sample means are shown in the table.

	0.5%	**1.0%**	**2.0%**	**2.5%**	**5.0%**	**95.0%**	**97.5%**	**98.0%**	**99.0%**	**99.5%**
Percentile	13.8	14.6	15.4	15.8	17.2	33.1	35.1	35.3	36.4	38.0

Selected percentiles from bootstrap distribution

a) What percentiles of the bootstrap distribution would be needed to create a 90% confidence interval?

b) Create a 90% confidence interval for the average penalty minutes for a player on the Ottawa Senators in the 2018–2019 season using the percentile method.

c) What percentiles of the bootstrap distribution would be needed to create a 98% confidence interval?

d) Create a 98% confidence interval for the average penalty minutes for a player on the Ottawa Senators in the 2018–2019 season using the percentile method.

e) Give an interpretation for one of the intervals (90% or 98%).

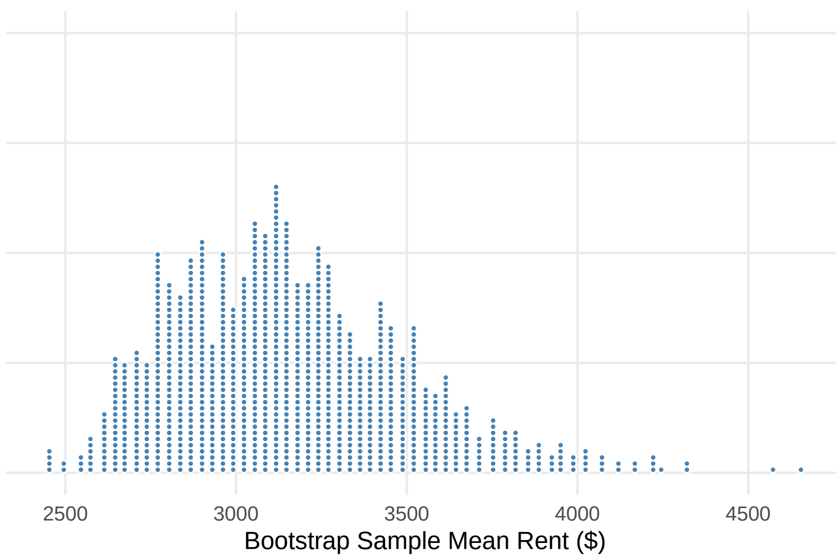
f) Using the 90% interval, is it plausible to say that the Ottawa Senators players were given more than 34 penalty minutes on average per player? Explain.

Class Example 2.3.11 *Manhattan apartment rent*

The New York City housing authority wants to know about the average rent for a one-bedroom apartment in Manhattan. One of their employees goes to Craigslist and records the monthly rent for a randomly selected sample of 8 listed one-bedroom apartments in Manhattan. Her sample is given below

3150 2275 3100 2650 2925 3305 2325 5495

Use the dotplot and the table below to answer the following questions.



	1.0%	2.5%	5.0%	10.0%	90.0%	95.0%	97.5%	99.0%
Percentile:	2595.25	2651.85	2720.40	2788.53	3546.80	3679.48	3796.40	3913.68

Selected percentiles from bootstrap distribution

a) Write down one possible bootstrap sample from the original sample.

b) Is it appropriate to use the bootstrap distribution to create a confidence interval for the average rent price in Manhattan? Justify your answer.

c) Regardless of how you answered b), create and interpret a 95% confidence interval for the average rent in Manhattan using the percentile method.

The *standard deviation* of the bootstrap distribution provides an estimate of the *standard error* for our statistic.

Class Example 2.3.12 *Skateboard Prices*

In a random sample of 20 skateboard prices on eBay, we find the average price is \$67.59. A bootstrap distribution gives a standard error of 10.9.

a) Find and interpret a 95% confidence interval.

b) Should a 90% confidence interval be wider or narrower than this interval?

c) If we took a sample of 50 prices instead of 20, would the 95% confidence interval be wider or narrower than this interval? Explain.

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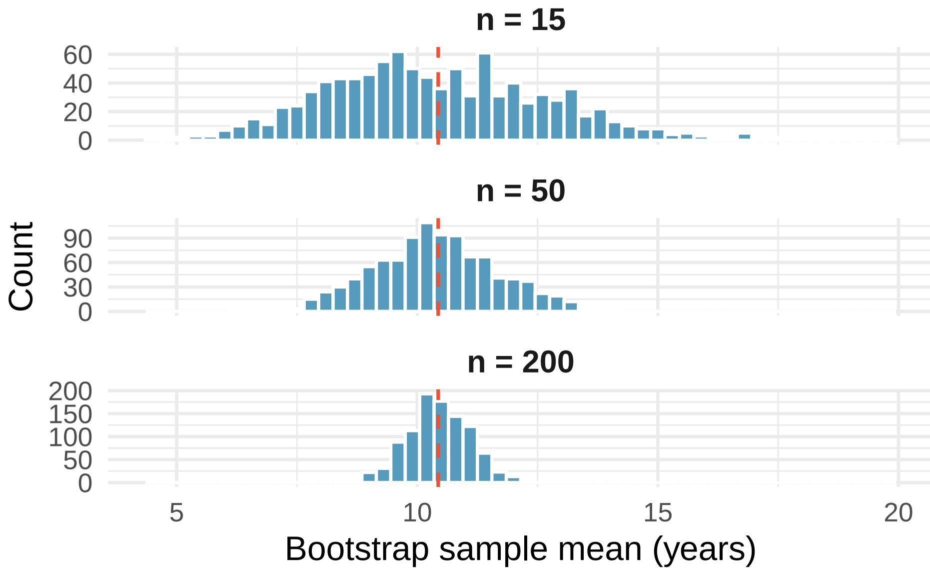

Figure 31: The effect of sample size on the bootstrap distribution. Larger samples produce narrower bootstrap distributions, leading to narrower confidence intervals. All three panels use the same population of penny ages.

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2.4

Key Concepts

- Explain the conclusion of the Central Limit Theorem (CLT)
- Recognize when the CLT can be applied to a study
- Identify when each of the three inference procedures should be used

The *Central Limit Theorem (CLT)* stats that when we take sufficiently large random samples from a population, the sampling distribution of many common statistics (such as $\hat{p}$ or $\overline{x}$) is approximately *normal* (bell-shaped), regarless of the shape of the underlying population distribution

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Class Example 2.4.1 *Visualizing the CLT*

Go to StatLens at https://learnlens.org/statlens/ and click on Sampling Distributions under the Simulate Tab. We will use the Right-Skew population shape. We wish to visualize the distribution of the sample mean.

a) Set the sample size to $n = 5$ and draw 1000 samples. What is the shape of the sampling distribution?

b) Set the sample size to $n = 30$ and draw 1000 samples. How does the shape of the sampling distribution change?

c) Set the sample size to $n = 100$ and draw 1000 samples. How does the shape of the sampling distribution change?

d) Which sample size would you use if you wanted the shape of the sampling distribution to be bell-shaped

e) Now, selected a Normal population shape. Set the sample size to $n = 5$ and draw 1000 samples. How does the shape of this sampling distribution compare to the sampling distribution shape from part (a)?

f) Set the sample size to $n = 30$ and draw 1000 samples. How does the shape of this sampling distribution change from part (e). How does it compare to the sampling distribution shape from part (b)?

Table 2: Comparison of inference methods

	Randomization	Bootstrap	Mathematical model
Question answered	Is there an effect? (hypothesis test)	How big is the parameter? (confidence interval)	Both
How variability is generated	Shuffle labels to simulate H_0	Resample with replacement from sample	Use formulas based on the normal distribution
What it models	Variability due to random assignment	Variability due to random sampling	Variability predicted by mathematical theory
Center of distribution	Null hypothesis value	Sample statistic	Depends on context (null values for tests, point estimates for CIs)
Conditions required	Minimal	Minimal	Independence + large sample
Computational cost	Moderate (need many simulations)	Moderate (need many resamples)	Low (just plug into formula)

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Class Example 2.4.2 *Applying the CLT*

For each of the following indicate whether the CLT can be applied. Why or why not?

a) We are interested in approximating the sampling distribution of the sample mean with samples of size 10.

b) We are interested in approximating the sampling distribution of the sample mean with samples of size 100.

c) We are interested in approximating the sampling distribution of the sample proportion when $n = 62$ and $\hat{p} = 0.05$.

d) We are interested in approximating the sampling distribution of the sample proportion when $n = 1000$ and $\hat{p} = 0.5$.

e) We are interested in approximating the sampling distribution of the sample minimum with samples of size 100.

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3.1

Key Concepts

- Estimate probabilities as areas under a density curve
- Recognize how the mean and standard deviation represent to the center and spread of a normal distribution
- Use the 68-95-99.7 rule to compute probabilities of intervals for normal distributions
- Use technology to compute probabilities of intervals for normal distributions
- Use technology to find endpoint(s) of intervals with a specified probability for normal distributions
- Convert in either direction between a general normal distribution, denoted $N(\mu, \sigma)$, and a standard normal distribution, denoted $N(0, 1)$

A *density curve* is a graphical representation of the distribution for a *continuous* random variable (*i.e.* sample space with probabilities).

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The *normal distribution* is a symmetric, bell-shaped distribution described by two parameters: the mean, μ (which determines the center), and the standard deviation, σ (which determines the spread). We write the distribution shorthand as $N(\mu, \sigma)$.

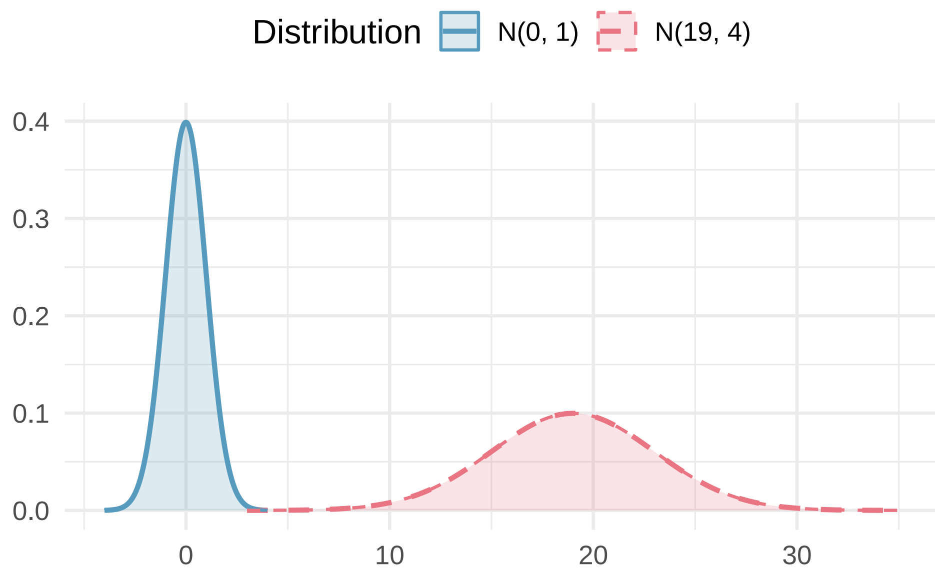

Figure 32: Two normal distributions with different parameters. The left curve has $\mu = 0$ and $\sigma = 1$; the right curve has $\mu = 19$ and $\sigma = 4$. Changing μ shifts the curve left or right; changing σ stretches or compresses it.

The *standard normal distribution* is denoted $N(0, 1)$.

The *standard score* or *z-score* indicates how many standard deviations a particular value is from the mean.

Class Example 3.1.1 *Standardizing to compare*

For each of the following, compare the results using their standardized score

a) Benny and Aaron are on the same track team. They have recently competed at a track and field day, and have each won their respective events. Benny won the high jump with a jump of 75 inches, and Aaron won the 100m dash with a time of 12 seconds. Let $J \sim N(65, 2)$ be the height of the jumps and $D \sim N(15, 1)$ be the times of the 100m dash. Who performed better Benny or Aaron?

b) A car manufacturer has a small car that gets 36 mpg and a truck that gets 22 mpg. Let $C \sim N(35, 4)$ be the distribution of mpgs for small cars and $T \sim N(19, 3)$ be the mpgs for trucks. Which vehicle performs better relative to the other vehicles in its class?

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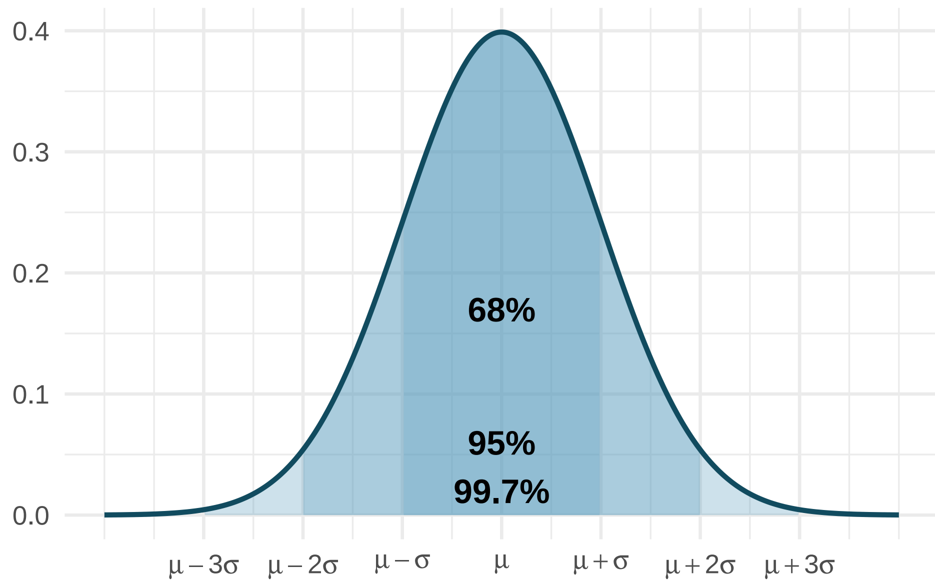

Figure 33: The 68-95-99.7 rule illustrated on a normal curve. The innermost shaded region (68%) extends from $\mu - \sigma$ to $\mu + \sigma$. The middle region (95%) extends from $\mu - 2\sigma$ to $\mu + 2\sigma$. The outermost region (99.7%) extends from $\mu - 3\sigma$ to $\mu + 3\sigma$.

Class Example 3.1.2 *SAT scores*

Let X be the SAT math scores for science and engineering majors, $X \sim N(609, 80)$. Using the 68-95-99.7 rule, answer the following questions.

a) What proportion of SAT scores should be between 529 and 689.

b) What is the probability that a randomly selected student with a science and engineering major has an SAT score of at least 769?

c) What proportion of SAT scores should be between 609 and 769?

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Class Example 3.1.3 *Rainfall in Ithaca*

Let R be the yearly rainfall (in inches) in Ithaca, NY. Data from Cornell's Northeast Regional Climate Center indicates that $R \sim N(35.4, 4.2)$.

a) How much rain should Ithaca expect to get in the year that is the 99^{th} percentile of rainfall?

b) How much rain should Ithaca expect to get in the year that marks the top 20 percent of rainfall?

c) What should the IQR be for yearly rainfall in Ithaca? (*i.e.* the middle 50%)

Class Example 3.1.4 *Furniture assembly*

A company that markets build-it-yourself furniture sells a computer desk that is advertised with the claim "less than an hour to assemble." Let T be the time (in hours) it takes to assemble the furniture. Through post-purchase surveys, it appears that $T \sim N(1.29, 0.43)$.

a) What percentage of people should be able to assemble the desk in less than an hour?

b) What is the probability that a randomly selected person can assemble the desk in between 1 and 1.5 hours?

c) The company would like to change the advertising claim to read "less than ______________ to assemble" so that 90% of people can assemble the desk in the time listed. What time should they use (as the cutoff for the bottom 90%)?

The *standard error (SE)* of a statistic is the standard deivation of its sampling distribution. It measures how much the statistic varies from sample to sample.

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Class Example 3.1.5 *Stents and stroke risk*

A study examined whether stents reduce stroke risk. The observed difference in 30-day stroke rates (treatment minus control) was $\hat{p}_T - \hat{p}_C = 0.090$, with $SE = 0.028$. Construct a 95% confidence interval.

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3.2

Key Concepts

- Find the mean and standard error for a distribution of sample proportions and difference in sample proportions
- Identify when a normal distribution is an appropriate model for a distribution of sample proportions
- Recognize when a two-sample z-interval for proportions is the appropriate statistical inference procedure
- State the conditions for the one-sample z-test for a single proportion and two-sample z-test for a difference in population proportions
- Use a normal distribution, when appropriate, to compute a confidence interval for a population proportion and difference in proportions between two populations
- Use a normal distribution, when appropriate, to test a hypothesis about a population proportion and for the difference in population proportions

To ensure that the sampling distribution of $\hat{p}$ is not too skewed for the normal approximation to work well, the *success-failure condition* requires that

- $np \geq 10$ and
- $n(1-p) \geq 10$.

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Class Example 3.2.1 *Use the CLT to estimate the standard error*
For each of the following, find $SE(\hat{p})$.

a) Samples of size $n = 100$ from a population with $p = 0.5$

b) Samples of size $n = 200$ from a population with $p = 0.5$

c) Samples of size $n = 500$ from a population with $p = 0.5$

d) What relationship did you notice from the answers from parts a) - c)?

Class Example 3.2.2 *Describe the sampling distribution*
For each of the following questions, identify if you can use the CLT. If so, fully describe the distribution. If not, identify the smallest sample size you need in order to use the CLT.

a) Samples of size $n = 10$ from a population with $p = 0.4$.

$\hat{p} \sim N($______________, ______________$)$ or Smallest n: ______________

b) Samples of size $n = 100$ from a population with $p = 0.6$.

$\hat{p} \sim N($________________ , ________________$)$ or Smallest n: ________________

c) Samples of size $n = 40$ from a population with $p = 0.9$.

$\hat{p} \sim N($________________ , ________________$)$ or Smallest n: ________________

Class Example 3.2.3 *Distributional Shape*
For each of the scenarios in Example 11.2, use StatLens to generate the Sampling Distribution for the Proportion using 3000 samples. What distributional shape is exhibited in each case? Do the cases that satisfy the "large sample" requirements look normal as the CLT suggest they shoul

a) Distributional shape:

b) Distributional shape:

c) Distributional shape:

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We can only use the CLT for confidence intervals if
-$n\hat{p} \geq 10$and -$n(1-\hat{p}) \geq 10$

Class Example 3.2.4 *Finding* z^*
Find the z^* values for the following confidence intervals.

a) 92%:

b) 85%:

c) 80%:

Class Example 3.2.5 *Left-handed desks*
A university would like to estimate the proportion of its students that are left-handed to purchase new desks for a building renovation. They take a random sample of 300 students and find that 32 people in the sample are left-handed.

a) Using the CLT find a 95% confidence interval for the proportion of students that are left-handed.

b) Provide an interpretation for your interval in a).

c) The Vice Chancellor for facilities decides that 15% of the desks they order should be left-handed. Use the confidence interval in a) to argue whether or not this will be an adequate proportion of left-handed desks.

Class Example 3.2.6 *Adopting a child in the US revisited*
A survey of 1000 randomly selected American adults conducted in January 2013 stated that "44% say its too hard to adopt a child in the US."

a) Using the CLT find a 99% confidence interval for the proportion of American adults who think it is too difficult to adopt.

b) Provide an interpretation for your interval in a).

c) Use the confidence interval in a) to argue whether or not a majority of American adults believe that it is too difficult to adopt.

d) How would this confidence interval change if the 44% figure was based on a sample size of 2000 adults instead of 1000?

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Class Example 3.2.7 *Sample size determination 1*
Consider Example 11.6. Using the 44% figure from the initial poll, how large of a sample size is needed to estimate the proportion of adults that say it is too hard to adopt a child in the US to within $\pm 2\%$ with 99% confidence.

Class Example 3.2.8 *Sample size determination 2*
A company produces plastic bottles. Because millions of bottles are produced each day, they are unable to

check every bottle for defects. Instead, they check a sample of bottles each day to estimate the population proportion of defectives. If no preliminary sample is taken, how large a sample of bottles is needed to obtain a 95% confidence interval that estimates the population proportion within $\pm 4\%$?

We can only use the CLT for a hypothesis test if

- $np_0 \geq 10$ and
- $n(1 - p_0) \geq 10$

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Class Example 3.2.9 *Infected Trees*
A ponderosa pine forest in Colorado has a pine beetle infestation. The beetles bore into a tree and carry a fungus that ultimately kills the tree. The forest can be treated and not eliminated if the proportion of infested trees in the forest is less than 20%. In a random sample of 180 trees, it was found that 26 trees were infested. Based on this data and a 5% level of significance, should the forest be treated?

a) Define the relevant parameter(s) and state the null and alternative hypotheses. Be sure to check any relevant conditions.

b) What is the test statistic.

c) Find the p-value, and give a sketch to illustrate your p-value.

d) State your decision and interpret your findings in the context of the study

Class Example 3.2.10 *Cereal example from Unit 2 revisited*
Recall the motivating example in Unit 2 where researchers were interested in whether children showed a preference for the cereal box that had cartoons on it. In the sample of 80 children, 52 chose the box with cartoons. Conduct a hypothesis test at the 5% level (include all steps) to determine if children prefer cereal with cartoon marketing.

Class Example 3.2.11 *Plato's* Republic*: Syllable Patterns*
Prose rhythm is characterized as the occurrence of five-syllable sequences in long passages of text. This characterization may be used to assess the similarity among passages of text and sometimes the identity of authors. Syllables are often categorized as long or short. On analyzing Plato's *Republic*, researchers found that 26.1% of the five-syllable sequences contained two short and three long syllables (Wishart and Leach, *Computer Studies of the Humanities and Verbal Behavior*, Vol.3, pp. 90-99). Suppose that Greek archaeologists have found an ancient manuscript dating back to Plato's time (about 427-347 B.C.E.). A random sample of 317 five-syllable sequences from the newly discovered manuscript showed that 61 contained two short and three long syllables. Do the data indicate that the population proportion of this type of five-syllable sequence is different from the text of Plato's *Republic*? You may either use a 95% confidence interval or hypothesis test (use $\alpha = 0.05$) to support your conclusion.

Class Example 3.2.12 *Proportion of Elderly, Florida and Texas*
From the 2010 US Census, we learn that 17.6% of the residents of Florida are 65 years old or older while 10.5% of the residents of Texas are 65 years old or older. If independent, random samples of 300 Florida residents and 200 Texas residents are selected and the difference in the proportion of elderly residents $(p_{FL} - p_{TX})$ is calculated:

a) Describe the shape of the sampling distribution of differences in sample proportions.

b) What is the mean of the distribution of differences in sample proportions.

c) Find the standard error of the distribution of differences in sample proportions.

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Class Example 3.2.13 *Mobile Connections to Libraries*
In a random sample of 2,252 Americans age 16 and older, 11% of the 1,059 men and 16% of the 1,193 women said they have accessed library services via a mobile device.

a) What must be *assumed* in order to construct a confidence interval for the difference in proportions?

b) Find and interpret a 95% confidence interval for the difference in proportions accessing libraries via mobile devices, between men and women.

c) Does your interval in (b) suggest that there is a difference in the proportions of men and women accessing libraries via mobile devices? Which population has a higher proportion?

Class Example 3.2.14 *Seat belt use*
We are interested in comparing the proportions of high school and college students in Wisconsin who always wear seat belts. In a random sample of 712 high school students, 615 say they always use seat belts. In a random sample of 641 college students, 578 always use seat belts. Compute at 95% confidence interval for the difference in the population proportions of high school and college students in WI who report always wearing their seat belt. Does this interval suggest that the population proportion is higher for either group?

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$H_0 : p_1 - p_2 = 0$ is equivalent to writing $H_0 : p_1 = p_2$.

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Class Example 3.2.15 *Accuracy of Lie Detectors*
Participants in a study to evaluate the accuracy of lie detectors were divided into two groups, with one group reading true material and the other group reading false material, while connected to a lie detector. The two way table indicates whether the participants were lying or telling the truth and also whether the lie detector indicated they were lying or not. We are interested in determining if there is a difference in the proportion of times the lie detector says the person is lying, depending on whether the person is lying or telling the truth.

	Detector says lying	**Detector says not**	**Total**
Person lying	31	17	48
Person not lying	27	21	48
Total	58	38	96

a) Are the conditions met for the two-sample test for proportions? Verify your answer.

b) Find the three sample proportions for the proportion of times the lie detector says the person is lying (the proportion for the lying people, the proportion for the truthful people, and the pooled proportion)

c) Test to see if there is a difference in the proportion of times the lie detector says the person is lying, depending on whether the person is lying or telling the truth. Use a 5% level of significance and show all details of the test.

Class Example 3.2.16 *Tagging Penguins*
A study was conducted to see if tagging penguins with metal tags harms them. In the study, 100 penguins were randomly assigned to receive a metal tag or (as a control group) an electronic tag. One of the variables studied is survival rate ten years after the penguins were tagged. The scientists observed that 20% of the 50 metal tagged penguins survived while 36% of the 50 electronically tagged penguins survived.

a) Are the conditions met for using the normal distribution? Verify your answer.

b) Define the two population proportions p_1 and p_2.

c) Test at $\alpha = 0.05$ to see if the survival rate is lower for all metal tagged penguins than for all electronically tagged penguins. Do metal tags appear to reduce survival rate in penguins?

3.3

Key Concepts

- Find the mean and standard error of sample means
- Distinguish between separate samples and paried data when comparing two means
- Recognize when a t-distribution is an appropriate model for a distribution of sample means, difference in means between two populations, and the mean of paired differences.
- Use a t-distribution, when appropriate, to compute a confidence interval for a population mean, difference in means between two populations, and the mean of paired differences.

We estimate the standard error of $\overline{x}$ with $\frac{s}{\sqrt{n}}$

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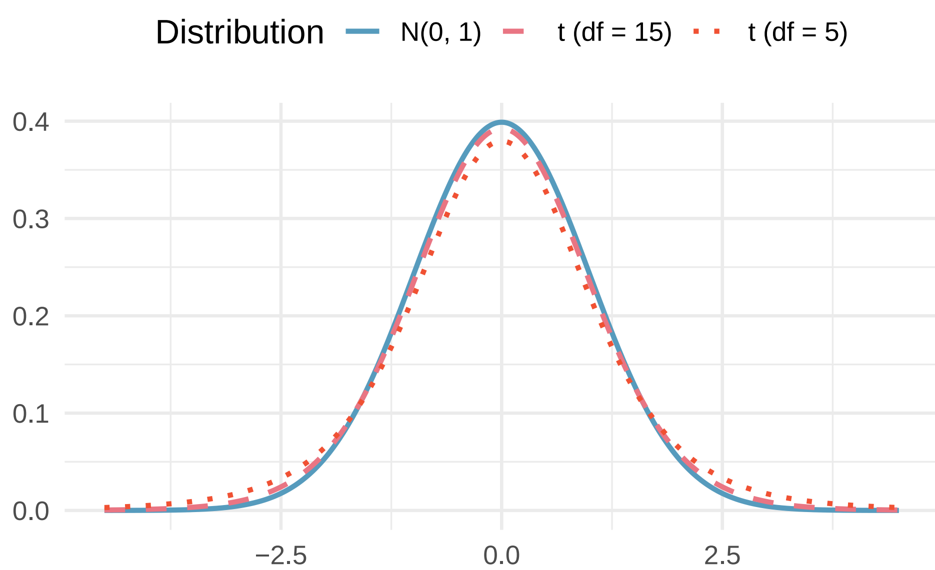

Figure 34: Three distributions plotted on the same axes: the standard normal distribution $N(0,1)$, a t-distribution with $df = 5$, and a t-distribution with $df = 15$. All are centered at 0 and bell-shaped, but the t-distributions have heavier tails, especially when df is small.

To use the t-distribution, you need either

- $n \geq 30$ or
- a population that is approximately normal

Class Example 3.3.1 *Describe the sampling distribution*
For each of the following questions, identify whether or not a t-distribution would be appropriate. If so, identify the number of degrees of freedom that you would use for the t-distribution.

a) Samples of size $n = 10$, from a population that is approximately normal.

b) Samples of size $n = 15$, from a population that is heavily skewed to the right.

c) Samples of size $n = 50$, from a population that is heavily skewed to the left.

We can only use the t-distribution for confidence intervals if

- $n \geq 30$ or
- the population is approximately normal

Class Example 3.3.2 *Triathlon Heart Rates*
The article "Cardiovascular and Thermal Response of Triathlon Performance" reports on a research study involving nine male triathletes. Maximum heart rate (beats/min) was recorded during performace of each of the three events. For swimming, the recorded values were 178, 182, 184, 184, 188, 190, 192, 196, 198.

a) What is the variable in this study? Is it categorical or quantitative?

b) What requirement is needed to make it appropriate to use the t confidence interval for this data?

c) Compute a 90% confidence interval for the mean heart rate of triathletes when swimming.

d) Provide an interpretation for your interval in c).

e) If we had calculated a 90% confidence interval instead of a 99% confidence interval, would it have been larger or smaller than the 99% confidence interval? Explain.

Class Example 3.3.3 *Dairy cows*
A study of 66 dairy cows found that the mean milk yield was 12.5 kg per milking with a standard deviation of 4.3 kg per milking.

a) Compute a 95% confidence interval for the average milk yield in the population. Make sure to check your conditions.

b) Provide an interpretation for your interval in a).

c) Is it plausible that the population mean milk yield is 11kg per milking?

d) If the mean and standard deviation were based off a sample fo 660 cows instead of 66, how would the 95% interval change?

We estimate the standard error of $\overline{x}_1 - \overline{x}_2$ with $\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

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Class Example 3.3.4 *Describe the sampling distribution*
For each of the following questions, identify whether or not a t-distribution would be appropriate.

a) Comparing averages from two samples of sizes $n_1 = 10$ and $n_2 = 15$, from populations that are approximately normal.

b) Comparing averages from two samples of sizes $n_1 = 15$ and $n_2 = 40$, where population 1 is heavily skewed right, and population 2 is approximately normal.

c) Comparing averages from two samples of sizes $n_1 = 50$ and $n_2 = 50$, from two populations that are heavily skewed to the left.

Remember, each sample must satisfy the necessary conditions for averages.

Class Example 3.3.5 *Highball vs tumbler*
Two common types of glassware at a bar are "highballs" (tall and thin) and "tumblers"(short and wide). Researchers randomly assigned participants either a "highball" glass or a "tumbler," each of which held 355 ml. Participants were asked to pour a shot (1.5 oz = 44.3 ml) into their glass. The summary statistics are given in the table below.

	n	$\overline{x}$	s
Highball	99	42.2 ml	16.2 ml
Tumbler	99	60.9 ml	17.9 ml

Summary statistics for amount poured

We want to know about the difference in the average amount of liquid poured into the two types of glasses.

a) What are the variables of interest in this study? Are they categorical or quantitative? Which is the explanatory variable and which is the response?

b) What is the parameter of interest?

c) Find and interpret a 95% confidence interval estimate for the difference in the average amount poured in the two different types of glasses. Use highball for group 1 and tumbler for group 2.

d) If you were the owner of a local bar, from a profit perspective, do you think there is any reason to prefer the highball to the tumbler glass?

Class Example 3.3.6 *Caffeine amounts*
A consumer agency wanted to estimate the difference in the amount of caffeine in two brands of coffee. The agency took a random sample of 15 one-pound jars of Brand I coffee, yielding a mean amount of caffeine of 80mg/jar with a standard deviation of 5mg. Another random sample of 12 jars of Brand II gave a eman of 77mg/jar with a standard deviation of 6mg. Find and interpret a 90% confidence interval for the difference in the population mean caffeine amounts. What does this interval tell us about the difference in caffiene for the two brands? Are there any conditions that must be satistified for the interval to be valid?

Class Example 3.3.7 *Deciding between difference in averages and paired difference*
For each of the following, identify the cases for the study and decide if it would be more appropriate to use a difference in averages or paired difference.

a) In a medical study to learn about the difference in the efficacy of generic versus brand name blood pressure medications, 50 individuals under 30 years old and 50 indiviudals over 30 years old were randomly assigned to receive either a generic medication or a brand name medication.

b) Researchers want to learn about the average age differences between partners who have been together more than 10 years. They randomly select 50 sets of partners and ask them their age.

c) In a study on the average difference in calories between servings of strawberry and vanilla yogurt, a food scientist randomly selected 12 brands of yogurt and recorded the calories in their strawberry and vanilla yogurts.

d) The Office of Greek Life randomly selects 45 fraternity and sorority members and looks to see if there is a difference in the average GPAs for fraternities vs sororities.

In this section, we will use a subscript $d$to indicate that we are working with differences.

We either need -$n_d \geq 30$or - the population of differences to be approximately normal

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Class Example 3.3.8 *Book prices*
You are given the task of determining if books differ in cost through Amazon.com or a local bookstore. The other student gathers data by selecting a random sample of 8 books from Amazon.com and a second independent random sample of 8 books from the local bookstore. The data are given below along with summary statistics.

Source									**Mean**	**SD**
Amazon	53	81	14	14	22	85	88	38	$\overline{x}_1 = 49.375$	$s_1 = 31.972$
Bookstore	45	76	43	64	21	94	47	25	$\overline{x}_2 = 51.875$	$s_2 = 24.868$

Summary statistics for book prices

a) Compute a 95% confidence interval based on how the other student collected data. What does this interval tell you about the difference in average price between Amazon and the local bookstore?

b) How might you have conducted this study instead? Why would a different study design be advantageous here?

c) Suppose that the data you collect from the different study design is given below. Based on the data that you collected, compute a 95% confidence interval for the difference in the cost of books for the two places? Does your conclusion differ from part a)? What accounted for this difference? (Hint: look at the SE for both cases)

Source								
Amazon	53	81	14	14	22	85	88	38
Bookstore	45	76	43	64	21	94	47	25

Summary statistics for book prices

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3.4

Key Concepts

- Distinguish between separate samples and paried data when comparing two means
- Recognize when a t-distribution is an appropriate model for a distribution of sample means, difference in means between two populations, and the mean of paired differences.
- Use a t-distribution, when appropriate, to test a hypothesis for a population mean, difference in means between two populations, and the mean of paired differences.

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Remember, t_{n-1} means a t-distribution with $n-1$ degrees of freedom

We can only use the t-distribution for confidence intervals if

- $n \geq 30$ or
- the population is approximately normal.

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Class Example 3.4.1 *How much is left?!*
Have you ever been frustrated because you could not get a container of some sort to release the last bit of its contents? The article "Shake, Rattle, and Squeeze: How Much is Left in That Container?" (*Consumer Reports*, May 2009: 8) reported on an investigation of this issue for various consumer products. In a random sample of five 6.0 ounce tubes of toothpaste from a particular brand and squeeze them until no more toothpaste will come out. Then each tube is cut open and the amount remaining is weighed. For this sample, the mean and standard deviation of the amount of toothpaste remaining (in ounces) was 0.502 and 0.102, respectively. Suppose the toothpaste manufacturer claims that it's packaging allows at least 90% of its contents to be released. Conduct a hypothesis test (using $\alpha = 0.05$) to test the manufacturer's claim. You may assume that the remaining amounts of toothpaste are approximately normally distributed.

a) Define the relevant parameter(s) and state the null and alternative hypotheses. Be sure to check any relevant conditions.

b) What is the test statistic.

c) Find the p-value and give a sketch to illustrate your p-value.

d) State your decision and interpret your findings in the context of the study.

Class Example 3.4.2 *Ski Patrol: Avalanches*
Snow avalanches can be a real problem for travelers in the western United States and Canada. A very common type of avalanche is called the slab avalanche. These have been studied extensively by David McClung, a professor of civil engineering at the University of British Columbia. Slab avalanches studied in Canada had an average thickness of 67 (in cm) (*Avalanche Handbook*, by David McClung and P. Schaerer.) The ski patrol at Vail, Colorado, is studying slab avalanches in their region. A random sample of avalanches in spring gave the following thicknesses (in cm).

59	**51**	**76**	**38**	**65**	**54**	**49**	**62**	**68**	**55**	**64**	**67**	**63**	**74**	**65**	**79**

Assuming that slab thickness has an approximately normal distribution, is there evidence that the slab thickness in the Vail region differs from that in Canada? You may either us a 95% confidence interval or a hypothesis test (use $\alpha = 0.05$) to support your conclusion.

Each sample must satisfy the necessary conditions for averages.

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Class Example 3.4.3 *Tire lifetimes*
A large automobile manufacturing company is trying to decide whether to purchase Goodyear or Firestone tires for its new models. They believe that Firestone tires might have a longer lifetime. To help arrive at a decision, an experiment is conducted where tires are run until they wear out. A random sample of 55 Goodyear tires gave a mean lifetime of 40,900 miles with a standard deviation of 3,200 miles. A random sample of 40 Firestone tires gave a mean lifetime of 42,800 miles and a standard deviation of 2,700 miles. Using a 1% discernable level, determine if Firestone tires have a larger population mean lifetime.

a) Define the relevant parameter(s) and state the null and alternative hypotheses. Be sure to check any relevant conditions.

b) What is the test statistic.

c) Find the p-value and give a sketch to illustrate your p-value.

d) State your decision and interpret your findings in the context of the study.

Class Example 3.4.4 *Caffeine amounts revisited*
The summary statistics from the amount of caffeine in coffee example are given in the table below.

	n	$\overline{x}$	s
Brand I	15	80mg	5mg
Brand II	12	77mg	6mg

Summary statistics for the amoung of caffeine in one-pound of coffee for two brands.

Conduct a hypothesis test to see if there is a difference between the average amount of caffiene for the two brands (use $\alpha = 0.05$). Do the findings from your hypothesis test agree with your confidence interval from Example (need ref num.)?

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Class Example 3.4.5 *Sales*
A company wants to know if attending a course on "how to be a successful salesperson" can increase the average sales of its employees. The company sent 6 of its salespeople to attend this course. The following gives the one-week sales of these employees before and after they attended the course. Determine if the course increases sales using a 5% significance level. What conditions/assumptions are necessary for the test to be valid?

Before	12	18	25	9	14	16
After	18	24	24	14	19	20

Sales of employees before and after they attended the course.

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4.1

Key Concepts

- Test a hypothesis about a categorical variable using a chi-square goodness-of-fit test
- Recognize when a chi-square distribution is appropriate for testing a categorical variable

The χ^2 goodness-of-fit test is used to test proportions when we have more than two groups in a single categorical variable.

Class Example 4.1.1 *Motivating dice example*
Suppose you are interested in determining if a die is fair, so you roll it 54 times and get 8 ones, 7 twos, 13 threes, 11 fours, 9 fives, and 6 sixes. What null and alternative hypothesis would you wante to use to determine if the die is fair?

The *expected count* for each category is found by multiplying the sample size by the hypothesized proportion for that category.

Class Example 4.1.2 *Motivating dice example continued*
If the null is true, and the die is fair, what would you expect for the number of ones, twos, threes, ect.?

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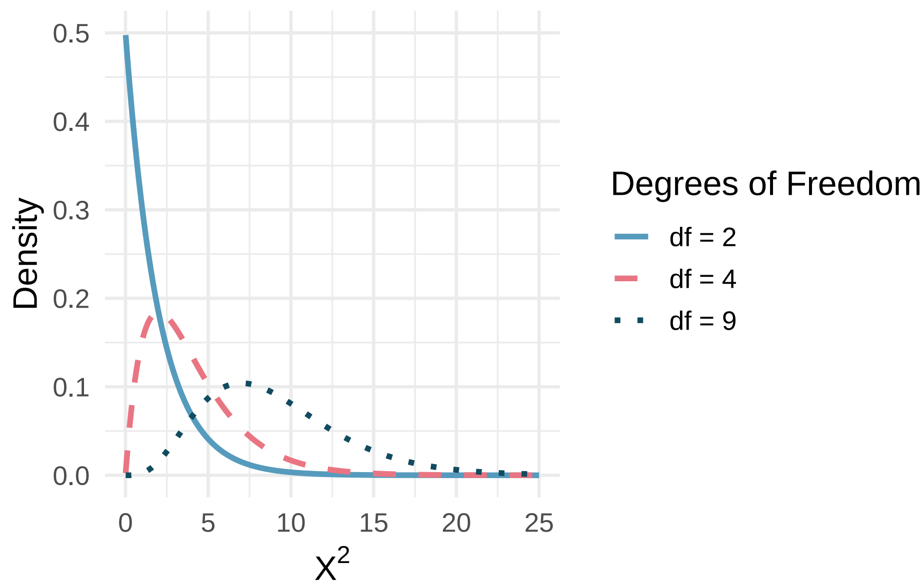

Figure 35: Three chi-square distributions with degrees of freedom 2, 4, and 9. As the degrees of freedom increase, the distribution becomes more symmetric, the center moves to the right, and the variability increases.

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Class Example 4.1.3 *Motivating dice example continued*
What are the test statistic and p-value for determineing if the die is fair? (Check conditions!) What is our conclusion?

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Class Example 4.1.4 *Distribution of fish*
The Game and Fish Department stocked a lake with fish in the following proportions: 30% catfish, 15% bass, 40% bluegill, and 15% pike. Five years later it sampled the lake to see if the distribution of the fish had changed. It found that the 500 fish in the sample were distributed as follows: 117 catfish, 88 bass, 223 bluegill, and 72 pike. Determine if the proportions of the types of fish in the lake have changed in the past 5 years by using a 5% significance level.

Class Example 4.1.5 *Age discrimination on U.S. juries*
It is often said that older people are overrepresented on juries in the United States. The *UCLA Law Review* randomly selected cases in one county in California, and recorded the number of jurors in various age groups given in the table below. Do the following data suggest that the *UCLA Law Review* should expand the scope of their study to more counties (they would do this if they observed discrimination)? Use $\alpha = 0.05$.

	21-40	**41-50**	**51-60**	**61+**	**Total**
Countywide Proportion (%)	42.0	23.0	16.0	19.0	100.0
Grand Jurors (counts)	10	18	38	66	132
Expected (counts)					

Actual ages of jurors in California county

4.2

Key Concepts

- Test a hypothesis about an association between two categorical variable using a chi-square association test
- Recognize when a chi-square distribution is appropriate for testing an association between two categorical variables

The χ^2 association test is used to test the relationship between two categorical variables

Class Example 4.2.1 *Traffic cameras and injuries*
Many cities have introduced traffic cameras at high-risk street intersections in an effort to reduce the severity of traffic accidents. Citizen rights groups have complained about the efficacy of these cameras. In fact, they doubt they have any impact whatsoever. A sample of 2938 reported accidents was collected including the type of accident (Death/Disabling Injury, Evident Injury, or Probable Injury) and camera status at the intersection (Before Camera, After Camera). What are the hypotheses to test the relationship between accident severity and presence of a traffic camera?

The expected counts come from the probability rules. If A and B are *independent*, then $P(A \cap B) = P(A) * P(B)$

Class Example 4.2.2 *Traffic cameras and injuries, continued*
The table below summarizes the traffic camera data. Use this data to compute a table of expected counts.

	Before Camera	**After Camera**
Death/Disabling	61	27
Evident	210	136
Probable	1659	845

Traffic camera status and accident severity

We want to see if the *observed* counts are far enough away from the *expected* counts

To use the χ^2 distribution, each expected count must be at least 5 or larger

Class Example 4.2.3 *Traffic cameras and injuries, continued*
Find the χ^2 test statistic and p-value using the traffic camera data to test if there is an association between traffic cameras and injury severity. (Check conditions!) What is our conclusion?

Class Example 4.2.4 *Star Trek*
In Star Trek, the colored uniforms worn by the crew identify their work area. Among Trekkies (fans of the show), they have begun to notice a pattern - they believe there is an association with uniform color and if the character suffers a fatality on the show. The table below gives a breakdown of the various crew colors and the number of fatalities - does this give evidence to suggest they are related?

	Blue	**Gold**	**Red**
Fatality	7	9	24
Survival	129	64	263

Uniform color and fatality count

4.3

Key Concepts

- Use ANOVA to test for a difference in means among several groups
- Explain how variation between groups and variation within groups are relevant for testing a difference in means between multiple groups.
- Create confidence intervals for single means and differences of means after doing an ANOVA for means, recognizing the problem of multiplicity

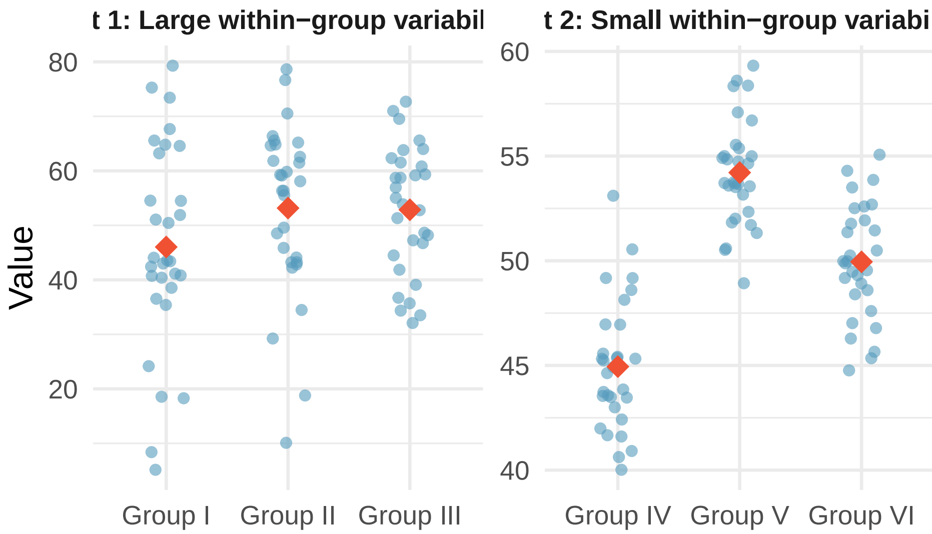

Figure 36: Side-by-side dot plots for two sets of three groups. In the first set (Groups I, II, III), the within-group variability is large relative to the between-group differences. In the second set (Groups IV, V, VI), the within-group variability is small and the between-group differences are clearly visible.

ANOVA is an extension of the two-sample t-test to more than two groups.

Class Example 4.3.1 *Quality control for lenses*
A company that produces glass lenses would like to know if the three presses they use are producing the same average thickness in lens. They hire an engineer to test to see if the machines should be calibrated. Write the hypothesis to test this using an ANOVA hypothesis test.

SS stands from *sum of sqaures*

MS stands from *mean sqaures*

B indicates "between" and W indicates "within"

Class Example 4.3.2 *Quality control continued*
Fill in the remaining spots in the ANOVA table below.

Source	**df**	**SS**	**MS**	**F**	**p-value**
Group		10.30			
Error					
Total	14	34.40			

Quality control ANOVA table

Find the p-value for the F-statistic that tests to see whether or not average glass thickness differs for each press. What are the degrees of freedom for the numerator and denominator? State your decision and provide an interpretation in the context of the problem.

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Class Example 4.3.3 *Caffeine and finger tapping*
Does caffeine consumption have an effect on twitch movements? To investigate this question, the following experiment was conducted. 30 subjects were randomly assigned to one of three groups, with 10 subjects in each group. The subjects were then given a tablet containing 0mg, 100mg, or 200mg of caffeine. Two hours later, they were asked to tap their finger as quickly as possible for one minute. Summary statistics are provided in the table below.

	0 mg	**100 mg**	**200 mg**
$\overline{x}$	244.8	246.4	248.3
s	2.394	2.066	2.214

Summary statistics for caffeine and finger tapping experiment

a) What are the variables recorded in this study? Are they categorical or quantitative? Which of the variables is the independent variable? Dependent variable?

b) Based on the summary statistics above, do you think that caffeine has an effect on finger tapping?

c) Test to see whether caffeine consumption causees a difference in the average number of finger taps (use $\alpha = 0.05$)

Source	**df**	**SS**	**MS**	**F**	p**-value**
Group		61.4			
Error		134.1			
Total					

Caffeine and finger tapping ANOVA table

f) Is there a *practical* difference between the groups? Is this the same as a *statistical* difference?

Pairwise comparisons is just a fancy way of saying consider all the possible pairs of means and conduct hypothesis tests to see if they are different.

Pairwise comparisons are also called *multiple comparisons* or *post-hoc tests*

Class Example 4.3.4 *Caffeine and finger tapping continues*
How many pairwise comparisons would we have? Suppose that there had been four caffeine levels instead. How many pairwise comparisons would this make?

The *family-wise error rate (FWER)* is the chance that we made a Type I error in at least one of our tests.

Class Example 4.3.5 *Caffeine and finger tapping continues*

IF we conduct pairwise comparisons for the average number of finger taps for the 3 caffeiene levels at $\alpha = 0.05$, what is the FWER?

Class Example 4.3.6 *Bonferroni correction for finger taps*
If we want the overall Type I error rate to be 0.05, what is the Bonferroni corrected α^* we should use when comparing the average number of finger taps for the 3 caffeine levels? What Bonferroni corrected α^* would we use if there were 4 caffeine levels instead of 3?

Class Example 4.3.7 *Restaurant tips*
In this example, we will be conducting many different tests from a dataset that contains information on the tipping patterns of patrons of the *First Crush* bistro in northern New York state. The data from 157 bills include the amount of the bill, size of the tip, percentage tip, number of customers in the group, whether or not a credit card was used, day of the week, and a coded identity of the server (A, B, or C). The first four variables are quantitative and the last three are categorical. For each test, you can use a hypothesis test or confidence

interval (if appropriate). On each test, be sure to verify the necessary conditions.

a) Most restaurant patrons leave a tip between 15% and 20% of the bill, and some restaurant customers determine the percent tip to leave based on the quality of the service. Summary statistics for mean tip percentage left for each of the three servers and a partial ANOVA table are given in the tables below.

Server	**Sample size**	**Mean**	**Std. Dev.**
A	60	17.543	5.504
B	65	16.017	3.485
C	32	16.109	3.376

Mean tip percentage by server

Source	**df**	**SS**	**MS**	**F**	**p-value**
Server			41.57		
Error					
Total		3001			

ANOVA table for mean tip percentage

i. Conduct the appropriate hypothesis test to see if there a difference in mean tip percentage between the three servers (use $\alpha = 0.05$).

ii. Is it appropriate to examine pairwise differences to see which of the servers differs in average tip percentage? Explain.

iii. Regardless of your answer in a.ii, if we want the overall Type I error rate to be 0.05, what is the Bonferroni corrected α we should use when comparing the average tip percentage for the servers?

c) The table below summarizes the number of bills by server. Conduct the appropriate hypothesis test to see if the servers serve equal numbers of tables.

Server	**A**	**B**	**C**	**Total**
Number of bills	60	65	32	157

Number of bills by server

4.4

Key Concepts

- Use the linear correlation coefficient, r, to quantify the strength and direction of a linear relationship
- Identify when the correlation coefficient can capture the trend between two quantitative variables
- Recognize that correlation does not equal causation
- Use computer output to make predictions and interpret coefficients using a fitted simple linear model
- Construct a confidence interval for the slope in a regression model
- Test a hypothesis about the slope of a regression model
- Test for evidence of a linear association between two quantitative variables using a sample correlation
- Compute and interpret the value of R^2 for a regression model
- Check a scatterplot for obvious departures from a simple linear model

A *scatterplot* is a graph of the relationship between two quantitative variables

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Class Example 4.4.1 *Describing Relationships*
Researchers captured 104 brushtail possums in Australia and took body measurements before releasing them. Consider the relationship between total body length (cm) and head length (mm).

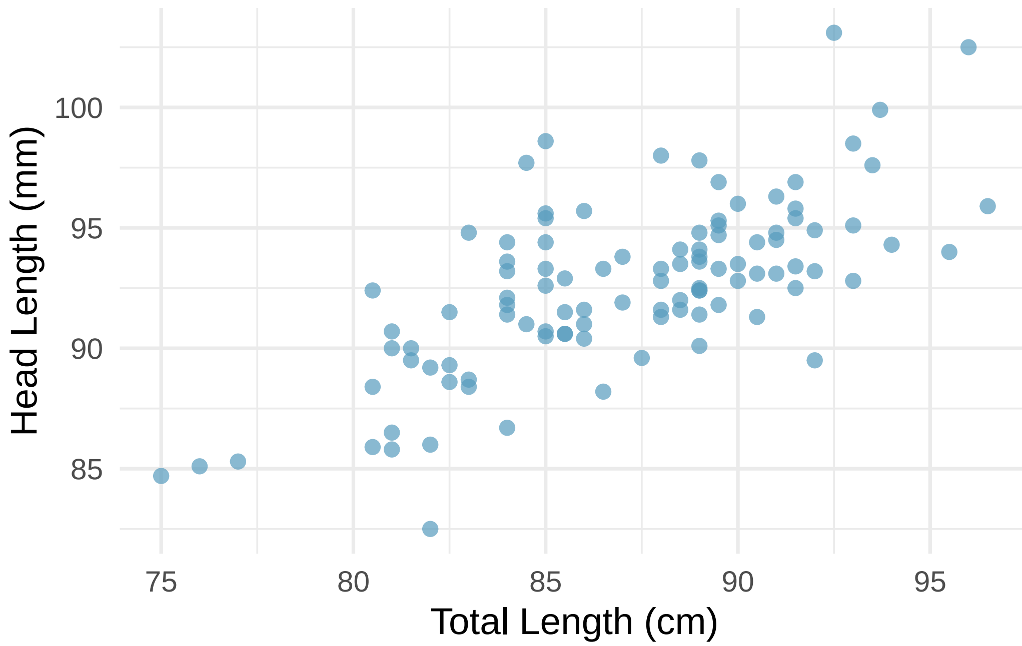

Figure 37: A scatterplot showing head length against total length for 104 brushtail possums. The relationship is moderately strong, positive, and linear.

Describe the relationship between total length and head length.

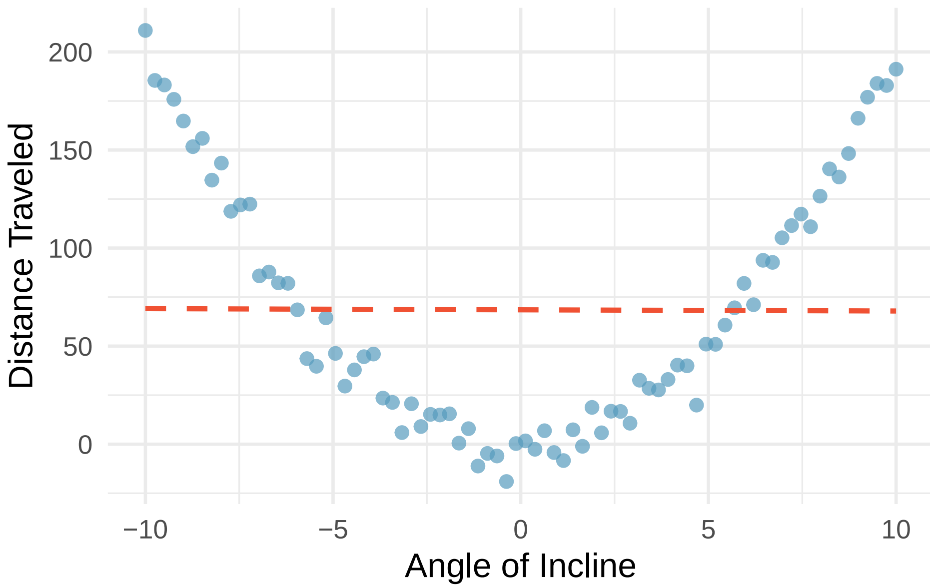

Figure 38: A scatterplot showing a clear quadratic (U-shaped) relationship between angle of incline and distance traveled. The best-fitting straight line is flat and does not capture the pattern at all.

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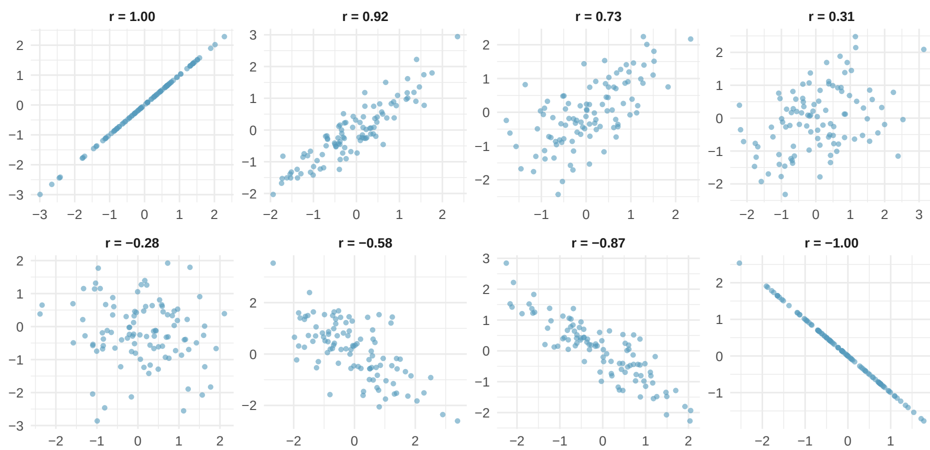

Figure 39: Eight scatterplots showing different correlation values. The first row shows positive relationships with r values of 1.00, 0.92, 0.73, and 0.31. The second row shows negative relationships with r values of -0.28, -0.58, -0.87, and -1.00.

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Class Example 4.4.2 *Firefighters and Damage*
Studies have found a positive correlation between the number of firefighters sent to a fire and the amount of damage the fire causes. Does this mean sending more firefighters causes more damage?

Notation:

- β_0: population intercept
- β_1: population slope
- ε: random error
- $\widehat{y}$: predicted value
- b_0: intercept estimate
- b_1: slope estimate

The *slope* and *correlation coefficient* should *always* have the same sign

Class Example 4.4.3 *Possums*
Researchers captured 104 brushtail possums in Australia and took body measurements. We consider total body length (cm) as the predictor to predict head length (mm). Some summary statistics and a scatterplot of the data are given below

Variable	**Average**	**Std. Dev.**
Body Length	87.09	4.31
Head Length	92.60	3.57
	$r = .691$	

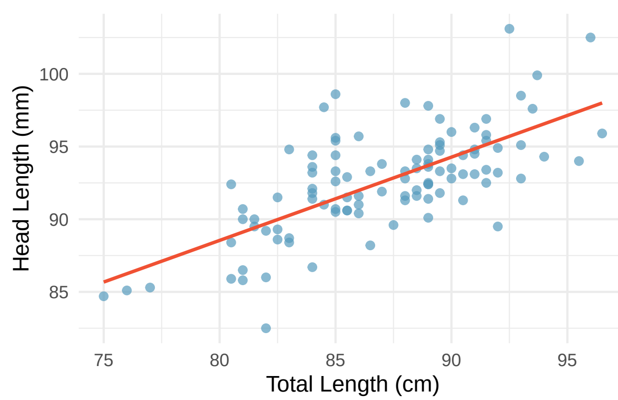

Figure 40: A scatterplot showing head length against total length for 104 brushtail possums, with a linear model fit to the data.

a) Looking at the plot, would you categorize the correlation as positive or negative? Strong, moderate, or weak?

b) Use the summaries to find estimates for the slope and intercept for the regression line. Record your answer in the form $\hat{y} = b_0 + b_1 x$.

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A *residual* is the difference between an observed values and a predicted value of a response

The *least squares line* is the equation that has the smallest sum of squared residuals

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Class Example 4.4.4 *Possums, revisited*

a) Predict the head length of a possum whose body length is 90mm. Is this prediction extrapolation?

b) One of the possums had a body length of 83mm and a head length of 88.4mm. Find the predicted value and residual for this possum.

c) Provide an interpretation of the slope and intercept (if reasonable) in the context of this problem.

SE_{b_1} is the standard error of b_1

Class Example 4.4.5 *Possums continued*
The table below gives the output from results for a simple linear regression of possum head length. Use the output to conduct a hypothesis test to see if there is some linear relationship between the average body length and average head length of possums, using $\alpha = 0.05$. Compute a 95% confidence interval for β_1. Does the results of the confidence interval match the results of the hypothesis test?

Coefficients	**Estimate**	**Std. Error**	t
(Intercept)	42.71	5.17	8.257
Length	0.57	0.06	?

Output from results for a simple linear regression of possum head lengths with body length

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R^2 gives the percentage of variation in the *response* variable that we can explain with the *explanatory* variable

When fitting a simple linear regression model, we can get the coefficient of determination by squaring the correlation, r.

Class Example 4.4.6 *Investigating the coefficient of determination*
Match each of the following scatterplots below (each displays the fitted model as the dashed line/curve) to one of the following values of coefficient of determination: $R^2 = .98$, $R^2 = .80$, $R^2 = .37$, or $R^2 = .001$

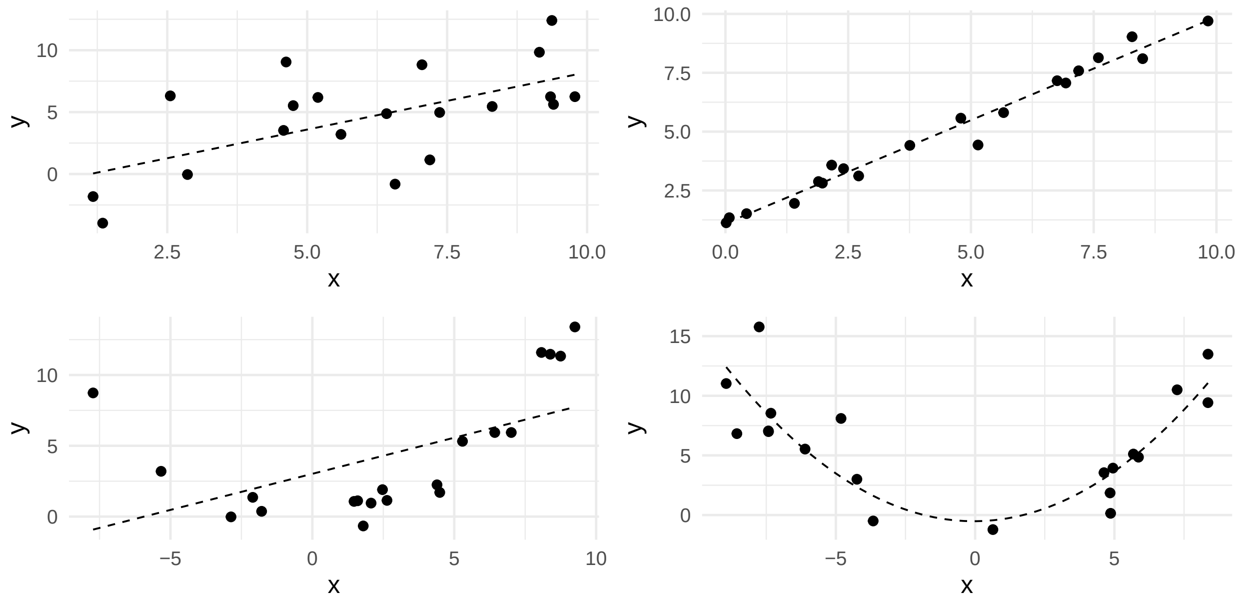

Figure 41: Four scatterplots with fitted regression model fit to the data.

Class Example 4.4.7 R^2 *for the Possum data*
If the correlation coefficient for the simple linear regression model relating the body length and head length of possums is 0.691, find and interpret the coefficient of determination, R^2.

Class Example 4.4.8 *Possum data, revisited*
In the previous examples, we use the variable *body length* to predict *head length*. Now, we are going to use *body length* to predict *skull width*. The following table give the output from a simple linear regression model of skull width with body length.

Coefficients	**Estimate**	**Std. Error**	t	p**-value (2-sided)**
(Intercept)	23.77	5.30	4.48	<0.0001
Length	0.38	0.06	?	<0.001
		$r = 0.526$		

Output from results for a simple linear regression of skull width with body length

a) Using the output above, write the regression equation in the form $\hat{y} = b_0 + b_1 x$. Use it to find the predicted and residual value for a possum with a length of 80.5mm and a skull width of 56mm.

b) Interpret the slope of the line in the context of the problem.

d) Conduct a hypothesis test to see if there is a significant positive relationship between length and skull width. Use $\alpha = 0.05$.

e) Find R^2 for this linear fit. Do body length explain head length or skull width better? Explain.

4.5

Key Concepts

- Use ANOVA to test the effectiveness of a simple linear model
- Compute and interpret the value of R^2 for a simple linear model using values in the ANOVA table.
- Find the standard deviation of the error term for a simple linear model
- Find the standard error of the slope for a simple linear model
- Use computer output to make predictions and interpret coefficients using a multiple regression model
- Use ANOVA to test the overall effectiveness of a multiple regression model
- Test the effectiveness of individual terms in a multiple regression model
- Compute and interpret the value of R^2 for a multiple regression model

Variability in the response variable can be partitioned into two components:

- $SS_{\text{Regression}}$
- SS_{Error}

Class Example 4.5.1 *Elmhurst College*
Elmhurst College conducted a study on the amount of gift aid given to students based on their family income (*The Chronicles of Higher Education*, 2012). 50 incoming freshman were randomly sampled and the amount of aid they were given along with their family's income were collected, both in thousands of dollars. A partially completed ANOVA table and other regression output are given below.

Source	**df**	**SS**	**MS**	**F**	**p-value**
Regression		363.16			
Error		1097.92			
Total					

Family income predicted gift aid ANOVA table

Coefficients	**Estimate**	**Std. Error**	t	**p-value (2-sided)**
(Intercept)	24.32	1.291	18.831	<0.0001
Income	-0.043	0.011	-3.985	0.0002

Output from results for a simple linear regression of gift aid with family income

a) Complete the partial ANOVA table and use it to calculate R^2 and provide an interpretation in the context of the study.

b) Conduct an F test to determine if the linear model fit is appropriate.

c) How does the F test statistic relate to the t test statistic provided in the output?

Class Example 4.5.2 *Elmhurst College, continued*
Calculate the standard error of the slope using the formula above and verify it matches what is in the output. Note that $s_x = 63.21$.

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Class Example 4.5.3 *Cherry Trees*
Timber yield is approximately equal to the volume of a tree, however, this value is difficult to measure without first cutting the tree down. Instead, other variables, such as height and diameter, may be used to predict a tree's volume and yield. Researchers wanting to understand the relationship between these variables for black cherry trees collected data from 31 such trees in the Allegheny National Forest, Pennsylvania. Height, is measured in feet, diameter in inches (at 54in above the ground), and volume in cubic feet (Hand, 1994). The multiple linear regression output obtained via Jamovi are provided below.

Model Fit Measures

Model	R	R^2
1	0.974	0.948

Note. Models estimated using sample size of N=31

Omnibus ANOVA Test

	Sum of Squares	df	Mean Square	F
diam	4782.974	1	4782.974	317.413
height	102.381	1	102.381	6.794
Residuals	421.921	28	15.069	

Note. Type 3 sum of squares

Model Coefficients - volume

Predictor	Estimate	SE	t	p
Intercept	–57.988	8.638	–6.713	<.001
diam	4.708	0.264	17.816	<.001
height	0.339	0.130	2.607	0.014

Figure 42: Jamovi output for multiple linear regression for the black cherry tree data.

a) Provide the estimated model equation relating height and diameter to the volume of the cherry trees.

b) Interpret each of the slope parameter estimates.

c) Give the coefficient of determination and interpret

d) Determine if at least one of height or diameter is a useful predictor of volume using an overall F test.

Class Example 4.5.4 *Cherry Trees, continued*
Using the Jamovi output, determine separately which predictors are needed in the multiple regression model. For each provide the null and alternative hypothesis, test statistic, p-value, and a brief statement as to whether or not the variable is needed.

Multicollinearity refers to when two (or more) of the predictors are associated with each other.

Class Example 4.5.5 *Cherry Trees, continued*
A scatterplot of the predictors height and diameter is provided below. Does this plot suggest multicollinearity is present? How does this impact the individual t tests done previously?

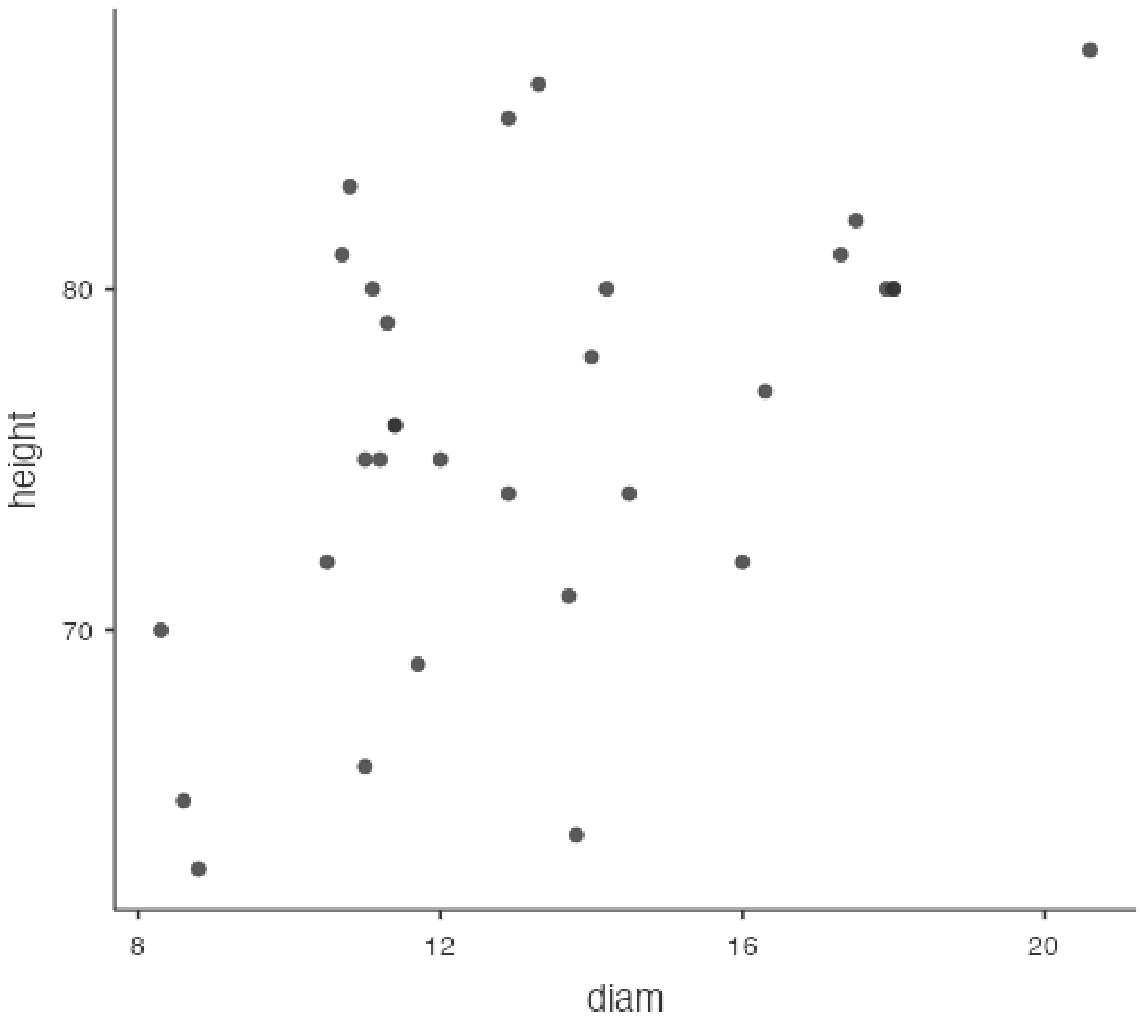

Figure 43: Scatterplot of the height and diameter of the cherry trees.

5.1

Key Concepts

- Compute the probability of events if outcomes are equally likely
- Identify when a probability question is asking for A and B, A or B, not A, or A if/given B
- Use the complement, additive, multiplicative, and conditional rules to compute probabilities of events
- Recognize when two events are disjoint

Class Example 5.1.1 *Roll a fair die*
A "die", the singular of "dice", is a cube with six faces numbered 1, 2, 3, 4, 5, and 6.

a) What is the chance of getting a 1 when rolling a fair die?

b) What is the chance of getting a 1 or 2 on the next roll?

c) What is the chance of not rolling a 2?

An *event* is a subset of the sample space, that is, a collection of one or more outcomes.

We use the notation $P()$ to mean *probability of*.

We use the notation $N()$ to mean *number of outcomes in* an event. When outcomes are equally likely, the probability of event is the number of outcomes in the event divided by the total number of outcomes.

Class Example 5.1.2 *Roll a fair die, continued*
Define the sample space for rolling a die.

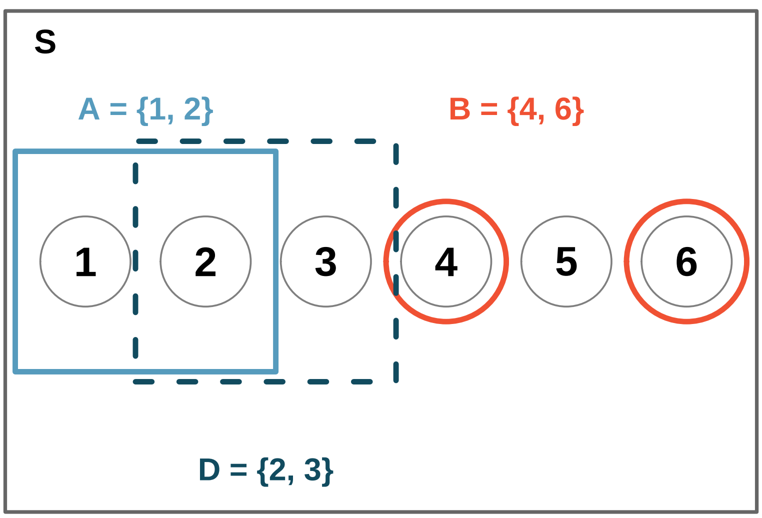

Figure 44: Three events for die outcomes. A and B are disjoint since they share no outcomes.

The *complement* of a condition, written "not A" or A^c, includes events in which the condition does not happen.

Class Example 5.1.3 *Roll a fair die, continued*

a) Write out the event for the complement of D and find its probability.

b) What is $P(D) + P(D^c)$?

Class Example 5.1.4 *Two rolls of a fair die*
Suppose a fair, six-sided die is rolled *twice*. Determine the following.
a) $N(\Omega)$

b) $N(A)$, where A is the event that the sum of the two rolls is 5

c) $P(B)$, where B is the event that the two rolls are the same

d) $P(C)$, where C is the event that the sum of the two rolls is even

e) $P(D)$, where D is the event that the two rolls are not the same

The *joint* or *intersection* of two conditions, written "A and B" or $A \cap B$, includes events that satisfy *both* conditions.

The *union* of two conditions, written "A or B" or $A \cup B$, includes events that satisfy *either or both* conditions.

The *conditional occurrence* of condition A given the occurrence of condition B is written "A if B" or $A \mid B$.

Events are *disjoint* if they cannot occur at the same time.

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Class Example 5.1.5 *Roll a fair die, revisited*
We are interested in the probability of rolling a 1, 4, or 5.

a) Explain why the outcomes 1, 4, and 5 are disjoint.

b) Apply the addition rule to determine $P(1 \text{ or } 4 \text{ or } 5)$.

Class Example 5.1.6 *Two rolls of a fair die, revisited*
We had the following events: A = sum of the two rolls is 5, B = the two rolls are the same, C = the sum of the two rolls is even.

a) Circle the pairs of events that are disjoint: A and B; A and C; B and C.

b) Find the probability of the union for all three pairs listed in part a).

Class Example 5.1.7 *Deck of Cards*
Consider a standard deck of 52 cards. In a standard deck, there are 2 colors: red and black. The red cards are split into two suits: diamonds and hearts. The black cards are split into clubs and spades. Each suit has 13 cards 2 through 10, jack, queen, king, ace. Let A be the event "the card is a diamond" and B "the card is a face card (jack, queen, or king)." Find the probability that a random card is a diamond or a face card.

Example 19.8 Types of classes
If 60% of college freshmen take a math class, 30% of college freshmen take a history class, and 12% of college freshmen take both a college math class and history class, what is the probability that a freshman selected at random is in a math or history class?

Example 19.9 La Crosse pets
Suppose 35% of households in La Crosse have a dog, 28% of households in La Crosse have a cat, and 52% of households in La Crosse have a dog or a cat. What is the probability that a La Crosse household selected at random has a dog and a cat?

The *conditional* occurrence of A given the occurrence of B is written "A if B" or "A given B" or A j B.

Example 19.10 Types of classes, revisited
If you know a selected student is in a math class, what is the probability they are in a history class?

Example 19.11 Baseball games
A Little League baseball team has a projected 80% chance of winning their first game in a tournament. If they win their first game, they are projected to win their second game with 60% probability. What is the probability that they win both their first and second game?

Class Example 5.1.8 *Flipping a fair coin*
Suppose we have a fair coin.

a) Suppose we flip the coin two times, are the flips independent of one another?

b) Suppose that the first three flips were HHT. What is the probability the 4^{th} flip is heads?

c) Suppose that the first three flips were TTT. What is the probability the 4^{th} flip is tails?

d) What is the probability of the event the 4 flips were HHTH?

e) What is the probability of the event the 4 flips were TTTT?

Example 19.13 Bin of marbles
A jar contains 20 blue marbles, 30 red marbles, and 50 yellow marbles. You select two marbles, one at a time, without looking. Find the following probabilities.

a) The probability that the second marble is red if the first marble was blue. You *have not* replaced the blue marble before drawing the second marble.

b) The probability that the second marble is red if the first marble was blue. You *have* replaced the blue marble before drawing the second marble.

c) The probability that the first marble is blue and the second marble is red. You *have not* replaced the blue marble before drawing the second marble.

d) The probability that the first marble is blue and the second marble is red. You *have* replaced the blue marble before drawing the second marble.

Class Example 5.1.9 *Relationship status and class year revisited*
Recall the following example from Unit 1. 169 college students were asked about relationship status and class year (lower=freshman/sophomore, upper=junior/senior). The results are given in the table below. For each of the following, assume that a student is chosen at random from this sample.

	Female	**Upper**	**Lower**
In a relationship	32	10	42
It's complicated	12	7	19
Single	63	45	108
Total	107	62	169

Contingency table for relationship status and class year.

a) What is the probability the student is in a relationship? Is this a joint, marginal, or conditional probability?

b) What is the probability the student is in a relationship given the student an upperclassmen? Is this a joint, marginal, or conditional probability?

c) What is the probability the student is an upperclassmen given the student is in a relationship? Is this a joint, marginal, or conditional probability?

d) What is the probability the student is a lowerclassmen and single? Is this a joint, marginal, or conditional probability?

The *basic counting rule* provides the number of possible outcomes of a random experiment in which several actions are performed.

A *combination* is any unordered collection elements chosen from a collection of elements.

Class Example 5.1.10 *True/False Quiz*
Suppose that a professor gives her students a 10 question True/False quiz.

a) How many ways are there to answer the quiz?

b) How many ways are there to answer exactly 5 questions correctly?

Class Example 5.1.11 *Cola Taste Test*
Suppose that three glasses are filled with the same cola, and then labeled C, D, and P. A randomly selected person is asked to order the three glasses by preference (this person is unaware that all three glasses contain the same cola).

a) How many ways can the cola preferences be ranked?

b) What is the probability that C is ranked first?

c) What is the probability that C is ranked first and D is ranked last?

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5.2

Key Concepts

- Identify a binomial random variable
- Compute probabilities for a binomial random variable
- Compute the mean and/or standard deviation for a binomial random variable

A *binomial random variable* measures the number of successes in a fixed number of trials.

1.
2.

3.

4.

Class Example 5.2.1 *Is it binomial?*
For each of the following, determine if the random variable is binomial. If it is, determine n and p.

a) On average, 25% of La Crosse residents recycle. 500 residents of La Crosse were asked if they recycle. A is the number of 'yes' answers in the sample.

b) A box contains 50 marbles, 35 green and 15 white. Fifteen marbles are selected without replacement. B is the number of white marbles in the sample.

c) An American roulette wheel consists of 18 red, 18 black, and 2 green numbers. Suppose you observed red on 10 consecutive spins. J is the number of additional spins needed until you observe black.

d) D is the number of 5s in ten rolls of a fair die.

e) Customers arrive at Alice's with a rate of 5 per hour. E is the number of customers that enter Alice's between 2 A.M. and 4 A.M.

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Class Example 5.2.2 *Testing ESP*
To test for ESP, we have 4 cards, each with different symbols. These cards are shuffled and one is chosen at random (symbol down). Each time, you guess the symbol on the card. Suppose that you repeat this test 8 times and you do not have ESP.

a) Let X be the number of times you guess correctly. Is X binomial? If so, what are n and p?

b) What is the expected value of X?

c) What is the standard deviation of X?

d) To be certified as having ESP, you need to correctly identify at least 6 cards of the 8 drawn. What is the probability you get certified?

Class Example 5.2.3 *Customer service*
Joan takes 10 phone calls per day from customers, and, on average, 30% of these callers are angry. If 7 or more of these callers in a given day are angry, Joan has a bad day.

a) Let C be the number of angry callers in a day. Is C binomial? If so, what are n and p?

b) How many angry callers will Joan speak to on average each day? What is the standard deviation for the number of angry callers?

c) What is the probability that Friday is a good day?

d) What is the probability that 3 of 5 days in her workweek are good?

Class Example 5.2.4 *Jordan Love*
In the 2023 NFL season, Jordan Love threw 579 passes. A fair-weather fan happened to only see a few games that season. The fan only saw Jordan Love throw 33 times, and of those 18 were completed (caught).

a) What would the fan estimate is Jordan Love's probability to complete a pass?

b) Based on the probability in part (a), how many passes would the fan expect Jordan Love to complete in the next 10 throws?

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5.3